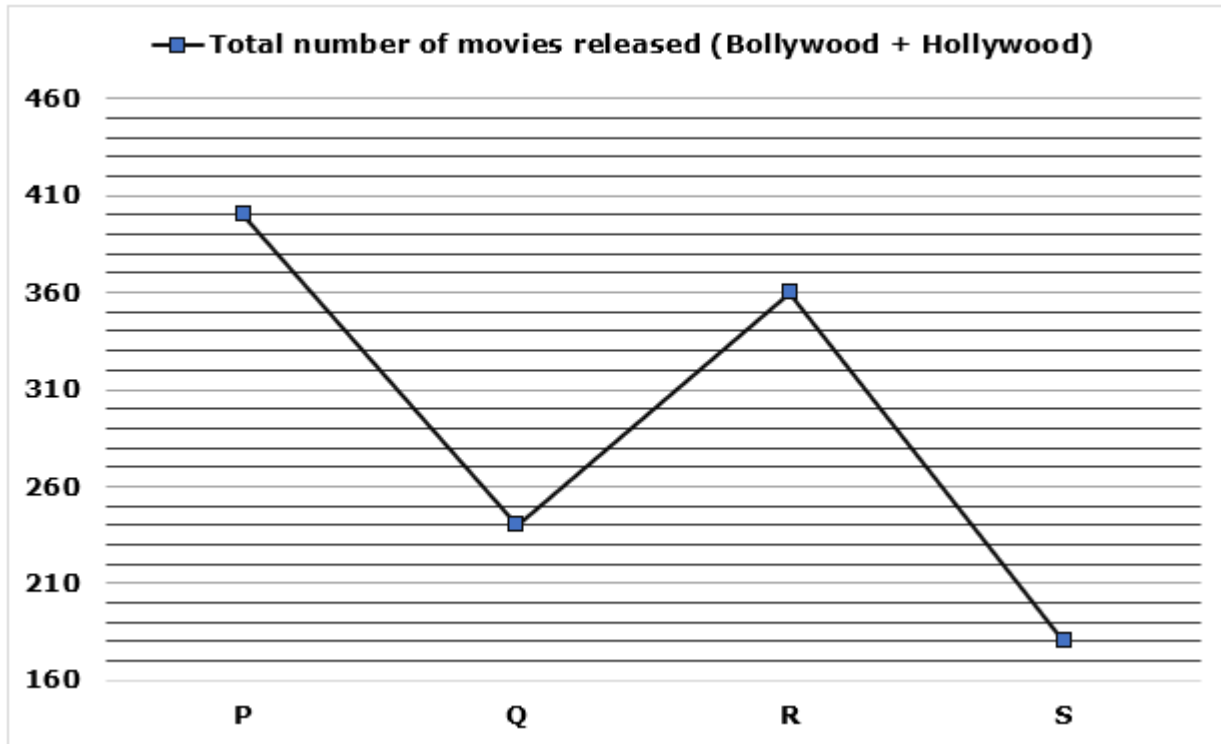


1. Questions

Read the following information carefully and answer the questions based on it.

The line graph given below shows the total number of movies released (Bollywood + Hollywood) in four theatres P, Q, R, and S. The table chart given below shows the total revenue generated (in Rs.) from all Bollywood and Hollywood movies released in each theatre.



Theatre	The total revenue generated from all Bollywood movies released (in Rs.)	The total revenue generated from all Hollywood movies released (in Rs.)
P	30000	15000
Q	9600	3200
R	32000	12800
S	5400	1350

Notes:

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- i). The total number of movies released = the number of Bollywood movies released + the number of Hollywood movies released
- ii). The total revenue generated = {(the total number of movies released) × (the price per ticket)}
- iii). In each theatre, the price per ticket of Bollywood and Hollywood movies may not be the same.
- iv). The ratio of the number of Bollywood movies released in theatre P and Q is 25:16.
- v). The price per ticket of Hollywood movies in theatre Q is 60% less than theatre P.
- vi). The ratio of the number of Hollywood movies released in theatre R and S is 32:9.

vii). The price per ticket of Bollywood movies in theatre S is $\frac{1}{4}^{\text{th}}$ of theatre R.

In theatre P, _____% of the total number of Bollywood movies are documentary, and the rest are non-documentary movies. If the number of non-documentary Hollywood movies is _____ less than the number of non-documentary Bollywood movies, then the ratio of the number of documentary Bollywood and Hollywood movies will be 3:2.

Which of the following does satisfy the filler(s) in the same order?

I: 60 and 50

II: 70 and 45

III: 80 and 30

- a. Only I follow
- b. Only II follow
- c. Both II and III follow
- d. All I, II and III follow
- e. None follow

2. Questions

In theatre S, if [D]% of discount is allowed on the price per ticket of Bollywood movies and the total number of Bollywood movies released is increased by [D+5], then the total revenue generated from all Bollywood movies released will be Rs. 5400. (D is a positive integer)

Which of the following can be determined?

I: The reduced price per ticket of Bollywood movies in theatre S after the discount is allowed.

II: The percentage increase in the total number of Bollywood movies released in theatre S.

III: The value of $[\sqrt{(D-6)}]$.

- a. Only I
- b. Only II
- c. Both II and III
- d. All I, II and III
- e. None can be determined

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3. Questions

In theatre Q, if the ratio of the number of tickets of Bollywood and Hollywood movies sold to females is 4:3 and the number of tickets of Bollywood movies sold to males is 70 more than the number of tickets of Hollywood movies sold to males, then find the difference between the number of tickets of Hollywood movies sold to males and females. (One person can purchase only one ticket)

- a. 20
- b. 28
- c. 16
- d. 12
- e. 10

4. Questions

If the number of Tollywood movies released in theatre S is $[3Y^2+5]$ and the price per ticket of Tollywood movies in theatre S is Rs. 75, then the total revenue generated from all Tollywood movies released in theatre S will be Rs. 600 more than the total revenue generated from all Bollywood movies released in the same theatre. Find the value of $[(Y/5)+1]$.

- a. 4
- b. 7
- c. 2
- d. 5
- e. None of these

5. Questions

If the total number of movies (Bollywood + Hollywood) released in another theatre T is half that of theatre P and the price per ticket of Bollywood and Hollywood movies in theatre T is Rs. 60 and Rs. 40, respectively, then the total revenue generated from all movies (Bollywood + Hollywood) released in theatre T will be Rs. 11000. Find the number of Hollywood movies released in theatre S is what percent of that of theatre T?

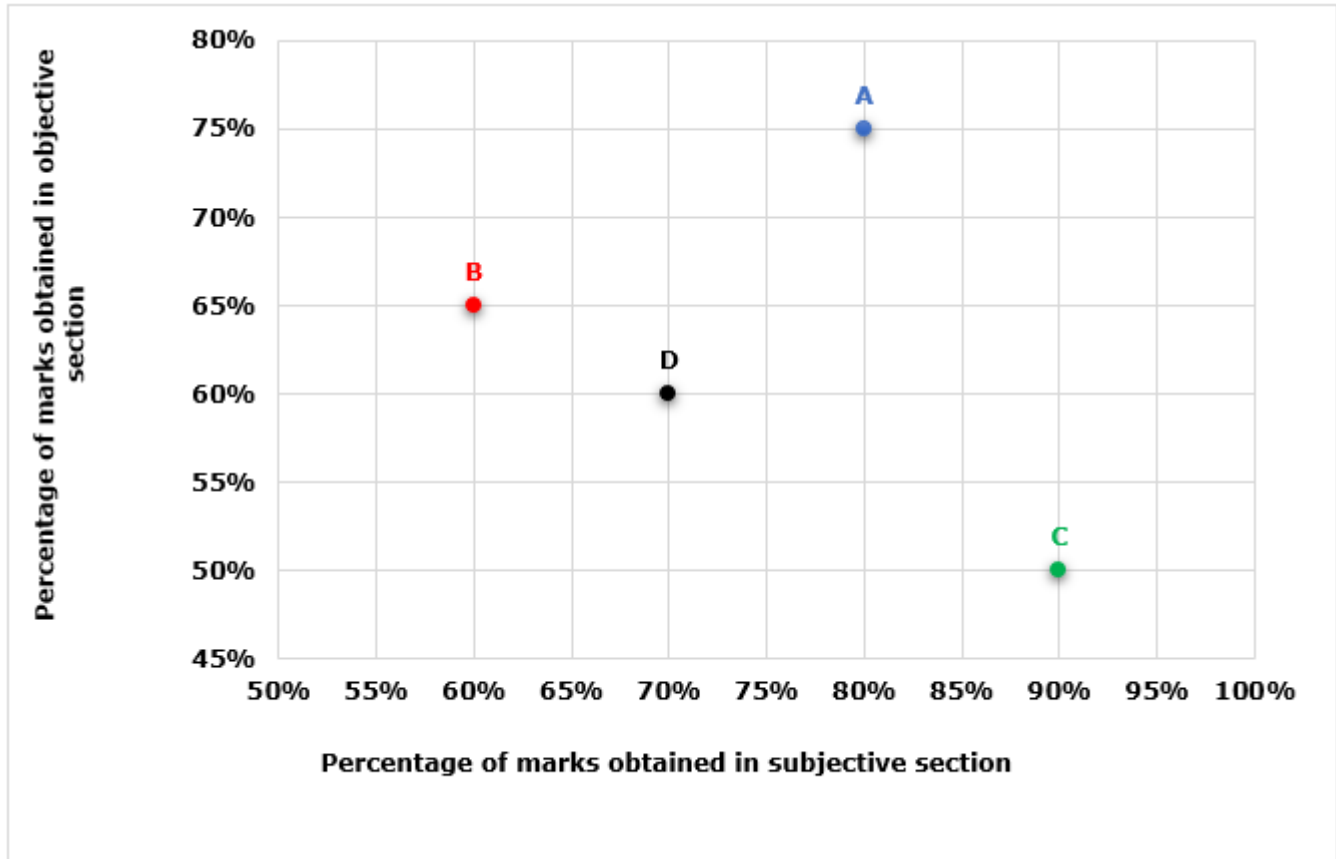
- a. 50%
- b. 75%
- c. 40%
- d. 65%
- e. None of these

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6. Questions

Read the following information carefully and answer the questions based on it.

The scatter graph given below shows the percentage of marks obtained by a student in the subjective and objective sections of four subjects A, B, C, and D.



Notes:

- i). The total number of sections in any subject = subjective section + objective section
- ii). In each section of each subject, the maximum marks and passing marks may not be the same.
- iii). The ratio of the maximum marks in the subjective and objective sections in subjects A, B, C, and D is 10:3, 15:4, 5:4, and 10:3, respectively.
- iv). The passing percentage of subjective sections in subjects A, B, C, and D is 40%, 30%, 20%, and 50%, respectively.
- v). The passing percentage of the objective section in subjects A, B, C, and D is 30%, 10%, 60%, and 40%, respectively.
- vi). In subject A, the marks obtained in the subjective section are 142 more than the passing marks in the objective section.
- vii). The difference between the marks obtained in the subjective and objective sections of subject B is 64. (subjective > objective)
- viii). The average of the passing marks in the subjective and objective sections of subject C is 34.
- ix). In subject D, the passing marks of the subjective section are 80 more than the marks obtained in the objective section.

If the student has attempted all questions contained in the subjective section of subject A, out of which some are correct, while the rest are wrong, then find the ratio of the number of correct and wrong attempts made by the student in the subjective section of subject A. (2 marks awarded for each correct attempt and 0.50 marks deducted for each wrong attempt)

- a. 19:6
- b. 17:8
- c. 24:1
- d. 21:4
- e. None of these

7. Questions

In subject B, $\frac{1}{3}$ rd and $\frac{1}{4}$ th of the maximum marks from the subjective and objective sections, respectively, are from part II, while the rest are from part I. If the percentage of marks obtained by the student in part I from the subjective and objective sections is 50% and 60%, respectively, then find the ratio of the marks obtained by the same student in part II from the subjective and objective sections in subject B.

- a. 5:1
- b. 3:2
- c. 4:3
- d. Cannot be determined
- e. None of these

8. Questions

In another subject E, the maximum marks of the subjective section are 300, and the student failed by $[X-20]$ marks in the same section. The passing marks in the subjective section of subject E are $[3X+30]$, while the student obtained $[X]\%$ of the maximum marks in the same section of the same subject.

Which of the following is/are definitely true about $[X]$?

I: Exactly divisible by 5 and 10.

II: Even composite number.

III: The sum of the digits is prime number.

- a. Only I
- b. Both I and II
- c. Both II and III
- d. All I, II and III
- e. None is true

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9. Questions

In the objective section of subject C, if 75% of the total number of questions contained are

attempted by the student, while the ratio of the number of correct and wrong attempts made by him is 11:4, then find the total number of correct attempts by the student in the objective section of subject C. (1 mark is awarded for each correct attempt, and 0.25 mark is deducted for each wrong attempt.)

- a. 15
- b. 10
- c. 44
- d. 25
- e. 16

10. Questions

In subject D, the ____[%]____ % of the maximum marks in subjective section are from practical, while the rest are from theory. In objective section, 30 marks are from practical, and the rest are from theory. The maximum mark in the theory of the objective section is 155 less than the subjective section.

What should come in place of the question mark?

- a. 50
- b. 20
- c. 10
- d. 40
- e. 30

11. Questions

Read the following information carefully and answer the questions based on it.

A certain number of students like two different activities, i.e., sports and arts, from three different schools, X, Y, and Z. Some students like only sports, some like only arts, while the rest like both sports and arts. Each student from each school likes at least one activity.

School X: The ratio of the total number of students who like sports and arts is 15:8. The number of students who like both sports and arts is 30. The number of students who like only sports is 140% more than the number of students who like only arts.

School Y: The ratio of the number of students who like only sports to the number of students who like only arts is 3:2. The number of students who like both sports and arts is 20 more than at school X. The total number of students who like arts is $\frac{3}{4}$ of the total number of students who like sports.

School Z: The number of students who like both sports and arts is 15. The number of students who like only sports is 85 more than the number of students who like only arts. The sum of the total number of students who like sports and the total number of students who like arts is 265.

Notes:

- i). The total number of students = the number of students who like only sports + the number of students who like only arts + the number of students who like both sports and arts.
- ii). The total number of students who like sports = the number of students who like only sports + the number of students who like both sports and arts.
- iii). The total number of students who like arts = the number of students who like only arts + the number of students who like both sports and arts.

In school X, out of the total number of students who likes sports, some likes cricket, while the rest likes football. Whereas, out of the total number of students who likes arts, some likes singing, while the rest likes dancing. If the ratio of the total number of students who likes cricket and dancing is 25:7, then the total number of students who likes football will be 2 less than singing. Find the difference between the total number of students who likes cricket and singing in school X.

- a. 36
- b. 48
- c. 24
- d. 12
- e. None of these

12. Questions

In school Y, if _____ % of the total number of students who like sports are boys and _____ % of the total number of students who like arts are girls, then the ratio of the number of girls who likes sports and boys who like arts will be 8:21.

Which of the following does satisfy the filler(s) in the same order?

I: 60 and 20

II: 80 and 30

III: 75 and 10

- a. Both I and II follow
- b. Both I and III follow
- c. Only II follow
- d. All I, II and III
- e. None follow

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13. Questions

In school Z, out of the total number of students who like sports, some are scholarship holders, which is $[2P + 5]$, while the rest are non-scholarship holders, which is 6 times that of scholarship holders; then the value of $[P]$ will be ____ $[?]$ ____.

What should come in place of question mark?

- a. 10
- b. 25
- c. 15
- d. 30
- e. None of these

14. Questions

In another school W, the number of students who like only sports is 50% more than the number of students who like only arts. If the number of students who like both sports and arts is 100 and the ratio of the total number of students who like sports and arts is 14:11, then find the total number of students in school X is what percentage of school W?

- a. 30%
- b. 60%
- c. 20%
- d. 50%
- e. 25%

15. Questions

In school X, out of the total number of students, the ratio of the number of boys and girls is [Q:R]. If the difference between the number of boys and girls in school X is 120, then find the value of [10Q% of 50R]. (boys > girls)

- a. 25
- b. 10
- c. 30
- d. 45
- e. None of these

16. Questions

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Read the following information carefully and answer the questions based on it.

The given table chart shows the percentage of the number of non-delivered orders out of the total number of orders placed (delivered + non-delivered) on five different days, i.e., Monday, Tuesday, Wednesday, Thursday, and Friday. It also shows the ratio of the number of delivered orders accepted and returned on each day.

Days	The percentage of the number of non-delivered orders	The ratio of the number of delivered orders accepted and returned
Monday	25%	5:1
Tuesday	20%	5:3
Wednesday	10%	3:2
Thursday	15%	11:6
Friday	30%	7:3

Note:

- i). The total number of orders placed = the number of orders delivered + the number of non-delivered orders
- ii). The total number of orders delivered = the number of delivered orders accepted + the number of delivered orders returned
- iii). The total number of orders placed on Tuesday is half of Thursday.
- iv). The number of delivered orders returned on Thursday is 20 more than the number of delivered orders accepted on Tuesday.
- v). The total number of orders placed on Monday is 20% more than Friday.
- vi). The number of delivered orders accepted on Friday is 102 more than the number of delivered orders returned on Monday.
- vii). On Wednesday, the number of non-delivered orders is 39 less than the number of returned delivered orders.

On Monday, if the total number of orders available (placed + non-placed) is $[40X^2 - 40]$, out of which $[60X]$ are non-placed orders, then the difference between the number of placed and non-placed orders on Monday will be $___[?]__$. (X is a positive integer)

What should come in place of the question mark?

- a. 140
- b. 120
- c. 100
- d. Cannot be determined
- e. None of these

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17. Questions

On Tuesday, if the number of electronic item orders accepted is 8 times the number of non-electronic orders items returned and the number of non-electronic item orders accepted is 40% of the number of electronic item orders returned, then find the total number of electronic items delivered (accepted + returned) on Tuesday. (Only electronic and non-electronic item orders are

delivered)

- a. 130
- b. 152
- c. 108
- d. 124
- e. None of these

18. Questions

On Wednesday, out of the total number of orders accepted, if the number of defective orders, which is equal to [Y] is further returned, then the new ratio of the number of orders accepted and returned on Wednesday will be 8:7.

Which of the following is/are definitely true about [Y]?

I: Perfect square.

II: Odd composite number.

III: Two-digit natural number.

- a. Both II and III
- b. Both I and II
- c. Only II
- d. All I, II and III
- e. None is true

19. Questions

If the number of orders delivered to males on Thursday is 90 more than Friday and the ratio of the number of orders delivered to females on Thursday and Friday is 7:5, then find the difference between the number of orders delivered to males and females on Thursday.

- a. 56
- b. 42
- c. 38
- d. 60
- e. 120

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20. Questions

The total number of orders placed on Saturday is 10% less than Tuesday, and the number of non-delivered orders on Saturday is [4Z]. If the ratio of the number of delivered orders accepted and returned on Saturday is 3:1, then the number of delivered orders returned on Saturday will be 20

more than the number of non-delivered orders on the same day. Find the value of (Z^2+5) .

- a. 41
- b. 69
- c. 30
- d. Cannot be determined
- e. 60

Explanations:

1. Questions

Theatre P and Q

The total number of movies released in theatre P = 400

The total number of movies released in theatre Q = 240

Let the number of Bollywood movies released in theatre P and Q be 25m and 16m respectively.

The number of Hollywood movies released in theatre P = $(400-25m)$

The number of Hollywood movies released in theatre Q = $(240-16m)$

So, the price per ticket of Hollywood movies in theatre P = Rs. $\{15000/(400-25m)\}$

And, the price per ticket of Hollywood movies in theatre Q = Rs. $\{3200/(240-16m)\}$

Now,

$$\{15000/(400-25m)\} / \{3200/(240-16m)\} = 100/40$$

$$\{15000/(400-25m)\} / \{3200/(240-16m)\} = 5/2$$

$$15/(400-25m) = 8/(240-16m)$$

$$3/(80-5m) = 1/(30-2m)$$

$$90-6m = 80-5m$$

$$m = 10$$

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Theatre R and S

The total number of movies released in theatre R = 360

The total number of movies released in theatre S = 180

Let the number of Hollywood movies released in theatre R and S be 32n and 9n respectively.

The number of Bollywood movies released in theatre R = $(360-32n)$

The number of Bollywood movies released in theatre S = $(180-9n)$

So, the price per ticket of Bollywood movies in theatre R = Rs. $\{32000/(360-32n)\}$

And, the price per ticket of Bollywood movies in theatre S = Rs. $\{5400/(180-9n)\}$

Now,

$$\{32000/(360-32n)\} / \{5400/(180-9n)\} = 4/1$$

$$\{40/(360-32n)\} / \{27/(180-9n)\} = 1/1$$

$$5/(45 - 4n) = 3/(20 - n)$$

$$100 - 5n = 135 - 12n$$

$$7n = 35$$

$$n = 5$$

Theatre P

The total number of movies released = 400

The total number of Bollywood movies released = $25m = (25 \times 10) = 250$

The total revenue generated from all Bollywood movies released = Rs. 30000

The price per ticket of Bollywood movies = $(30000/250) = \text{Rs. } 120$

The total number of Hollywood movies released = $(400-25m) = (400-250) = 150$

The total revenue generated from all Hollywood movies released = Rs. 15000

The price per ticket of Hollywood movies = $(15000/150) = \text{Rs. } 100$

Theatre Q

The total number of movies released = 240

The total number of Bollywood movies released = $16m = (16 \times 10) = 160$

The total revenue generated from all Bollywood movies released = Rs. 9600

The price per ticket of Bollywood movies = $(9600/160) = \text{Rs. } 60$

The total number of Hollywood movies released = $(240-16m) = (240-160) = 80$

The total revenue generated from all Hollywood movies released = Rs. 3200

The price per ticket of Hollywood movies = $(3200/80) = \text{Rs. } 40$

Theatre R

The total number of movies released = 360

The total number of Bollywood movies released = $(360-32n) = (360-32 \times 5) = (360-160) = 200$

The total revenue generated from all Bollywood movies released = Rs. 32000

The price per ticket of Bollywood movies = $(32000/200) = \text{Rs. } 160$

The total number of Hollywood movies released = $32n = 160$

The total revenue generated from all Hollywood movies released = Rs. 12800

The price per ticket of Hollywood movies = $(12800/160) = \text{Rs. } 80$

Theatre S

The total number of movies released = 180

The total number of Bollywood movies released = $(180-9n) = (180-9 \times 5) = (180-45) = 135$

The total revenue generated from all Bollywood movies released = Rs. 5400

The price per ticket of Bollywood movies = $(5400/135) = \text{Rs. } 40$

The total number of Hollywood movies released = $9n = 45$

The total revenue generated from all Hollywood movies released = Rs. 1350

The price per ticket of Hollywood movies = $(1350/45) = \text{Rs. } 30$

Theatre	The number of Bollywood movies released	The number of Hollywood movies released	The price per ticket of Bollywood movies	The price per ticket of Hollywood movies
P	250	150	120	100
Q	160	80	60	40
R	200	160	160	80
S	135	45	40	30

Answer: A

Now, according to the question,

Theatre P

The total number of Bollywood movies released = 250

The total number of Hollywood movies released = 150

Using I

The number of documentary Bollywood movies = 60% of 250 = 150

The number of non-documentary Bollywood movies = $(250-150) = 100$

The number of non-documentary Hollywood movies = $(100-50) = 50$

The number of documentary Hollywood movies = $(150-50) = 100$

So, the ratio of the number of documentary Bollywood and Hollywood movies = $150:100 = 3:2$.

Therefore, I do satisfy.

Using II

The number of documentary Bollywood movies = 70% of 250 = 175

The number of non-documentary Bollywood movies = $(250-175) = 75$

The number of non-documentary Hollywood movies = $(75-45) = 30$

The number of documentary Hollywood movies = $(150-30) = 120$

So, the ratio of the number of documentary Bollywood and Hollywood movies = $175:120 = 35:24$.

Therefore, II does not satisfy.

Using III

The number of documentary Bollywood movies = 80% of $250 = 200$

The number of non-documentary Bollywood movies = $(250-200) = 50$

The number of non-documentary Hollywood movies = $(50-30) = 20$

The number of documentary Hollywood movies = $(150-20) = 130$

So, the ratio of the number of documentary Bollywood and Hollywood movies = $200:130 = 20:13$.

Therefore, III does not satisfy.

So, only I follow.

Hence, the correct answer is option A.

2. Questions

Theatre P and Q

The total number of movies released in theatre P = 400

The total number of movies released in theatre Q = 240

Let the number of Bollywood movies released in theatre P and Q be $25m$ and $16m$ respectively.

The number of Hollywood movies released in theatre P = $(400-25m)$

The number of Hollywood movies released in theatre Q = $(240-16m)$

So, the price per ticket of Hollywood movies in theatre P = Rs. $\{15000/(400-25m)\}$

And, the price per ticket of Hollywood movies in theatre Q = Rs. $\{3200/(240-16m)\}$

Now,

$$\{15000/(400-25m)\} / \{3200/(240-16m)\} = 100/40$$

$$\{15000/(400-25m)\} / \{3200/(240-16m)\} = 5/2$$

$$15/(400-25m) = 8/(240-16m)$$

$$3/(80-5m) = 1/(30-2m)$$

$$90-6m = 80-5m$$

$$m = 10$$

Theatre R and S

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The total number of movies released in theatre R = 360

The total number of movies released in theatre S = 180

Let the number of Hollywood movies released in theatre R and S be $32n$ and $9n$ respectively.

The number of Bollywood movies released in theatre R = $(360 - 32n)$

The number of Bollywood movies released in theatre S = $(180 - 9n)$

So, the price per ticket of Bollywood movies in theatre R = Rs. $\{32000/(360 - 32n)\}$

And, the price per ticket of Bollywood movies in theatre S = Rs. $\{5400/(180 - 9n)\}$

Now,

$$\{32000/(360 - 32n)\} / \{5400/(180 - 9n)\} = 4/1$$

$$\{40/(360 - 32n)\} / \{27/(180 - 9n)\} = 1/1$$

$$5/(45 - 4n) = 3/(20 - n)$$

$$100 - 5n = 135 - 12n$$

$$7n = 35$$

$$n = 5$$

Theatre P

The total number of movies released = 400

The total number of Bollywood movies released = $25m = (25 \times 10) = 250$

The total revenue generated from all Bollywood movies released = Rs. 30000

The price per ticket of Bollywood movies = $(30000/250) = \text{Rs. } 120$

The total number of Hollywood movies released = $(400 - 25m) = (400 - 250) = 150$

The total revenue generated from all Hollywood movies released = Rs. 15000

The price per ticket of Hollywood movies = $(15000/150) = \text{Rs. } 100$

Theatre Q

The total number of movies released = 240

The total number of Bollywood movies released = $16m = (16 \times 10) = 160$

The total revenue generated from all Bollywood movies released = Rs. 9600

The price per ticket of Bollywood movies = $(9600/160) = \text{Rs. } 60$

The total number of Hollywood movies released = $(240 - 16m) = (240 - 160) = 80$

The total revenue generated from all Hollywood movies released = Rs. 3200

The price per ticket of Hollywood movies = $(3200/80) = \text{Rs. } 40$

Theatre R

The total number of movies released = 360

The total number of Bollywood movies released = $(360-32n) = (360-32 \times 5) = (360-160) = 200$

The total revenue generated from all Bollywood movies released = Rs. 32000

The price per ticket of Bollywood movies = $(32000/200) = \text{Rs. } 160$

The total number of Hollywood movies released = $32n = 160$

The total revenue generated from all Hollywood movies released = Rs. 12800

The price per ticket of Hollywood movies = $(12800/160) = \text{Rs. } 80$

Theatre S

The total number of movies released = 180

The total number of Bollywood movies released = $(180-9n) = (180-9 \times 5) = (180-45) = 135$

The total revenue generated from all Bollywood movies released = Rs. 5400

The price per ticket of Bollywood movies = $(5400/135) = \text{Rs. } 40$

The total number of Hollywood movies released = $9n = 45$

The total revenue generated from all Hollywood movies released = Rs. 1350

The price per ticket of Hollywood movies = $(1350/45) = \text{Rs. } 30$

Theatre	The number of Bollywood movies released	The number of Hollywood movies released	The price per ticket of Bollywood movies	The price per ticket of Hollywood movies
P	250	150	120	100
Q	160	80	60	40
R	200	160	160	80
S	135	45	40	30

Answer: D

Theatre S (Bollywood Movies)

The reduced price of per ticket after discount = $\{40-(D\% \text{ of } 40)\} = \text{Rs. } (40-0.40D)$

The increased number of Bollywood tickets sold = $\{135+(D \times 5)\} = (140+D)$

So, the total revenue generated from all Bollywood movies released = $\{(40-0.40D) \times (140+D)\} = \text{Rs. } (5600+40D-56D-0.40D^2)$

Now,

$$(5600+40D-56D-0.40D^2) = 5400$$

$$0.40D^2 + 56D - 40D - 200 = 0$$

$$0.40D^2 + 16D - 200 = 0$$

$$4D^2 + 160D - 2000 = 0$$

$$D^2 + 40D - 500 = 0$$

$$D^2 + 50D - 10D - 500 = 0$$

$$D(D+50) - 10(D+50) = 0$$

$$(D-10)(D+50) = 0$$

$$D = +10 \text{ or } -50$$

So, $D = 10$ (positive integer)

I: The reduced price of per ticket of Bollywood movies in theatre S after the discount is allowed =
 $(40 - 0.40D) = (40 - 0.40 \times 10) = (40 - 4) = \text{Rs. } 36$

So, I can be determined.

II: The initial number of Bollywood movies released in theatre S = 135

The increased number of Bollywood movies released in theatre S = $(140 + D) = (140 + 10) = 150$

Therefore, the required percentage increase = $[(150 - 135)/135] \times 100\% = 11.11\%$.

So, II can be determined.

III: The value of $\sqrt{D-6} = \sqrt{10-6} = \sqrt{4} = 2$.

So, III can be determined.

Therefore, all I, II and III can be determined.

Hence, the correct answer is option D.

3. Questions

Theatre P and Q

The total number of movies released in theatre P = 400

The total number of movies released in theatre Q = 240

Let the number of Bollywood movies released in theatre P and Q be 25m and 16m respectively.

The number of Hollywood movies released in theatre P = $(400 - 25m)$

The number of Hollywood movies released in theatre Q = $(240 - 16m)$

So, the price per ticket of Hollywood movies in theatre P = Rs. $\{15000/(400 - 25m)\}$

And, the price per ticket of Hollywood movies in theatre Q = Rs. $\{3200/(240 - 16m)\}$

Now,

$$\{15000/(400 - 25m)\} / \{3200/(240 - 16m)\} = 100/40$$

$$\{15000/(400-25m)\}/\{3200/(240-16m)\} = 5/2$$

$$15/(400-25m) = 8/(240-16m)$$

$$3/(80 - 5m) = 1/(30 - 2m)$$

$$90 - 6m = 80 - 5m$$

$$m = 10$$

Theatre R and S

The total number of movies released in theatre R = 360

The total number of movies released in theatre S = 180

Let the number of Hollywood movies released in theatre R and S be $32n$ and $9n$ respectively.

The number of Bollywood movies released in theatre R = $(360-32n)$

The number of Bollywood movies released in theatre S = $(180-9n)$

So, the price per ticket of Bollywood movies in theatre R = Rs. $\{32000/(360-32n)\}$

And, the price per ticket of Bollywood movies in theatre S = Rs. $\{5400/(180-9n)\}$

Now,

$$\{32000/(360-32n)\}/\{5400/(180-9n)\} = 4/1$$

$$\{40/(360-32n)\}/\{27/(180-9n)\} = 1/1$$

$$5/(45 - 4n) = 3/(20 - n)$$

$$100 - 5n = 135 - 12n$$

$$7n = 35$$

$$n = 5$$

Theatre P

The total number of movies released = 400

The total number of Bollywood movies released = $25m = (25 \times 10) = 250$

The total revenue generated from all Bollywood movies released = Rs. 30000

The price per ticket of Bollywood movies = $(30000/250) = \text{Rs. } 120$

The total number of Hollywood movies released = $(400-25m) = (400-250) = 150$

The total revenue generated from all Hollywood movies released = Rs. 15000

The price per ticket of Hollywood movies = $(15000/150) = \text{Rs. } 100$

Theatre Q

The total number of movies released = 240

The total number of Bollywood movies released = $16m = (16 \times 10) = 160$

The total revenue generated from all Bollywood movies released = Rs. 9600

The price per ticket of Bollywood movies = $(9600/160) = \text{Rs. } 60$

The total number of Hollywood movies released = $(240-16m) = (240-160) = 80$

The total revenue generated from all Hollywood movies released = Rs. 3200

The price per ticket of Hollywood movies = $(3200/80) = \text{Rs. } 40$

Theatre R

The total number of movies released = 360

The total number of Bollywood movies released = $(360-32n) = (360-32 \times 5) = (360-160) = 200$

The total revenue generated from all Bollywood movies released = Rs. 32000

The price per ticket of Bollywood movies = $(32000/200) = \text{Rs. } 160$

The total number of Hollywood movies released = $32n = 160$

The total revenue generated from all Hollywood movies released = Rs. 12800

The price per ticket of Hollywood movies = $(12800/160) = \text{Rs. } 80$

Theatre S

The total number of movies released = 180

The total number of Bollywood movies released = $(180-9n) = (180-9 \times 5) = (180-45) = 135$

The total revenue generated from all Bollywood movies released = Rs. 5400

The price per ticket of Bollywood movies = $(5400/135) = \text{Rs. } 40$

The total number of Hollywood movies released = $9n = 45$

The total revenue generated from all Hollywood movies released = Rs. 1350

The price per ticket of Hollywood movies = $(1350/45) = \text{Rs. } 30$

Theatre	The number of Bollywood movies released	The number of Hollywood movies released	The price per ticket of Bollywood movies	The price per ticket of Hollywood movies
P	250	150	120	100
Q	160	80	60	40
R	200	160	160	80
S	135	45	40	30

Answer: A

Theatre Q

The total number of tickets sold of Bollywood movies (males + females) = 160

The total number of tickets sold of Hollywood movies (males + females) = 80

Now, let the number of tickets of Bollywood and Hollywood movies sold to females be $4k$ and $3k$ respectively.

So, the number of tickets of Bollywood movies sold to males = $(160-4k)$

And, the number of tickets of Hollywood movies sold to males = $(80-3k)$

Now,

$$(160-4k) - (80-3k) = 70$$

$$90 - 4k = 80 - 3k$$

$$k = 10$$

So, the number of tickets of Hollywood movies sold to males = $(80-3k) = (80-3 \times 10) = (80-30) = 50$

And, the number of tickets of Hollywood movies sold to females = $3k = 30$

Therefore, the required difference = $(50-30) = 20$.

Hence, the correct answer is option A.

4. Questions

Theatre P and Q

The total number of movies released in theatre P = 400

The total number of movies released in theatre Q = 240

Let the number of Bollywood movies released in theatre P and Q be $25m$ and $16m$ respectively.

The number of Hollywood movies released in theatre P = $(400-25m)$

The number of Hollywood movies released in theatre Q = $(240-16m)$

So, the price per ticket of Hollywood movies in theatre P = Rs. $\{15000/(400-25m)\}$

And, the price per ticket of Hollywood movies in theatre Q = Rs. $\{3200/(240-16m)\}$

Now,

$$\{15000/(400-25m)\} / \{3200/(240-16m)\} = 100/40$$

$$\{15000/(400-25m)\} / \{3200/(240-16m)\} = 5/2$$

$$15/(400-25m) = 8/(240-16m)$$

$$3/(80 - 5m) = 1/(30 - 2m)$$

$$90 - 6m = 80 - 5m$$

$$m = 10$$

Theatre R and S

The total number of movies released in theatre R = 360

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The total number of movies released in theatre S = 180

Let the number of Hollywood movies released in theatre R and S be $32n$ and $9n$ respectively.

The number of Bollywood movies released in theatre R = $(360-32n)$

The number of Bollywood movies released in theatre S = $(180-9n)$

So, the price per ticket of Bollywood movies in theatre R = Rs. $\{32000/(360-32n)\}$

And, the price per ticket of Bollywood movies in theatre S = Rs. $\{5400/(180-9n)\}$

Now,

$$\{32000/(360-32n)\} / \{5400/(180-9n)\} = 4/1$$

$$\{40/(360-32n)\} / \{27/(180-9n)\} = 1/1$$

$$5/(45 - 4n) = 3/(20 - n)$$

$$100 - 5n = 135 - 12n$$

$$7n = 35$$

$$n = 5$$

Theatre P

The total number of movies released = 400

The total number of Bollywood movies released = $25m = (25 \times 10) = 250$

The total revenue generated from all Bollywood movies released = Rs. 30000

The price per ticket of Bollywood movies = $(30000/250) = \text{Rs. } 120$

The total number of Hollywood movies released = $(400-25m) = (400-250) = 150$

The total revenue generated from all Hollywood movies released = Rs. 15000

The price per ticket of Hollywood movies = $(15000/150) = \text{Rs. } 100$

Theatre Q

The total number of movies released = 240

The total number of Bollywood movies released = $16m = (16 \times 10) = 160$

The total revenue generated from all Bollywood movies released = Rs. 9600

The price per ticket of Bollywood movies = $(9600/160) = \text{Rs. } 60$

The total number of Hollywood movies released = $(240-16m) = (240-160) = 80$

The total revenue generated from all Hollywood movies released = Rs. 3200

The price per ticket of Hollywood movies = $(3200/80) = \text{Rs. } 40$

Theatre R

The total number of movies released = 360

The total number of Bollywood movies released = $(360-32n) = (360-32 \times 5) = (360-160) = 200$

The total revenue generated from all Bollywood movies released = Rs. 32000

The price per ticket of Bollywood movies = $(32000/200) = \text{Rs. } 160$

The total number of Hollywood movies released = $32n = 160$

The total revenue generated from all Hollywood movies released = Rs. 12800

The price per ticket of Hollywood movies = $(12800/160) = \text{Rs. } 80$

Theatre S

The total number of movies released = 180

The total number of Bollywood movies released = $(180-9n) = (180-9 \times 5) = (180-45) = 135$

The total revenue generated from all Bollywood movies released = Rs. 5400

The price per ticket of Bollywood movies = $(5400/135) = \text{Rs. } 40$

The total number of Hollywood movies released = $9n = 45$

The total revenue generated from all Hollywood movies released = Rs. 1350

The price per ticket of Hollywood movies = $(1350/45) = \text{Rs. } 30$

Theatre	The number of Bollywood movies released	The number of Hollywood movies released	The price per ticket of Bollywood movies	The price per ticket of Hollywood movies
P	250	150	120	100
Q	160	80	60	40
R	200	160	160	80
S	135	45	40	30

Answer: C

Theatre R (Tollywood Movies)

The total number of movies released = $(3Y^2+5)$

The price per ticket = Rs. 75

So, the total revenue generated = $\{(3Y^2+5) \times 75\} = \text{Rs. } (225Y^2+375)$

Now,

$$225Y^2 + 375 = 5400 + 600$$

$$225Y^2 = 6000 - 375$$

$$225Y^2 = 5625$$

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$$Y^2 = 25$$

$$Y = \sqrt{25}$$

$$Y = 5$$

Therefore, the value of $\{(Y/5)+1\} = \{(5/5)+1\} = (1+1) = 2$.

Hence, the correct answer is option C.

5. Questions

Theatre P and Q

The total number of movies released in theatre P = 400

The total number of movies released in theatre Q = 240

Let the number of Bollywood movies released in theatre P and Q be 25m and 16m respectively.

The number of Hollywood movies released in theatre P = $(400-25m)$

The number of Hollywood movies released in theatre Q = $(240-16m)$

So, the price per ticket of Hollywood movies in theatre P = Rs. $\{15000/(400-25m)\}$

And, the price per ticket of Hollywood movies in theatre Q = Rs. $\{3200/(240-16m)\}$

Now,

$$\{15000/(400-25m)\} / \{3200/(240-16m)\} = 100/40$$

$$\{15000/(400-25m)\} / \{3200/(240-16m)\} = 5/2$$

$$15/(400-25m) = 8/(240-16m)$$

$$3/(80 - 5m) = 1/(30 - 2m)$$

$$90 - 6m = 80 - 5m$$

$$m = 10$$

Theatre R and S

The total number of movies released in theatre R = 360

The total number of movies released in theatre S = 180

Let the number of Hollywood movies released in theatre R and S be $32n$ and $9n$ respectively.

The number of Bollywood movies released in theatre R = $(360-32n)$

The number of Bollywood movies released in theatre S = $(180-9n)$

So, the price per ticket of Bollywood movies in theatre R = Rs. $\{32000/(360-32n)\}$

And, the price per ticket of Bollywood movies in theatre S = Rs. $\{5400/(180-9n)\}$

Now,

$$\{32000/(360-32n)\} / \{5400/(180-9n)\} = 4/1$$

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$$\{40/(360-32n)\}/\{27/(180-9n)\} = 1/1$$

$$5/(45 - 4n) = 3/(20 - n)$$

$$100 - 5n = 135 - 12n$$

$$7n = 35$$

$$n = 5$$

Theatre P

The total number of movies released = 400

The total number of Bollywood movies released = $25m = (25 \times 10) = 250$

The total revenue generated from all Bollywood movies released = Rs. 30000

The price per ticket of Bollywood movies = $(30000/250) = \text{Rs. } 120$

The total number of Hollywood movies released = $(400-25m) = (400-250) = 150$

The total revenue generated from all Hollywood movies released = Rs. 15000

The price per ticket of Hollywood movies = $(15000/150) = \text{Rs. } 100$

Theatre Q

The total number of movies released = 240

The total number of Bollywood movies released = $16m = (16 \times 10) = 160$

The total revenue generated from all Bollywood movies released = Rs. 9600

The price per ticket of Bollywood movies = $(9600/160) = \text{Rs. } 60$

The total number of Hollywood movies released = $(240-16m) = (240-160) = 80$

The total revenue generated from all Hollywood movies released = Rs. 3200

The price per ticket of Hollywood movies = $(3200/80) = \text{Rs. } 40$

Theatre R

The total number of movies released = 360

The total number of Bollywood movies released = $(360-32n) = (360-32 \times 5) = (360-160) = 200$

The total revenue generated from all Bollywood movies released = Rs. 32000

The price per ticket of Bollywood movies = $(32000/200) = \text{Rs. } 160$

The total number of Hollywood movies released = $32n = 160$

The total revenue generated from all Hollywood movies released = Rs. 12800

The price per ticket of Hollywood movies = $(12800/160) = \text{Rs. } 80$

Theatre S

The total number of movies released = 180

The total number of Bollywood movies released = $(180 - 9n) = (180 - 9 \times 5) = (180 - 45) = 135$

The total revenue generated from all Bollywood movies released = Rs. 5400

The price per ticket of Bollywood movies = $(5400/135) = \text{Rs. } 40$

The total number of Hollywood movies released = $9n = 45$

The total revenue generated from all Hollywood movies released = Rs. 1350

The price per ticket of Hollywood movies = $(1350/45)$ = Rs. 30

Theatre	The number of Bollywood movies released	The number of Hollywood movies released	The price per ticket of Bollywood movies	The price per ticket of Hollywood movies
P	250	150	120	100
Q	160	80	60	40
R	200	160	160	80
S	135	45	40	30

Answer: E

Theatre T

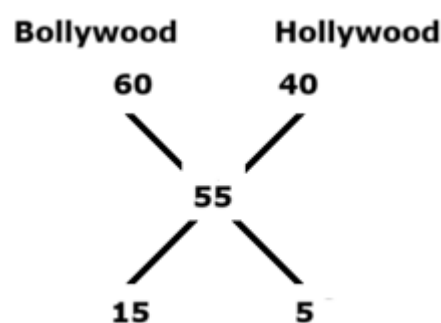
The total number of movies (Bollywood + Hollywood) released = $(400/2) = 200$

The price per ticket of Bollywood movies = Rs. 60

The price per ticket of Hollywood movies = Rs. 40

The overall price per ticket of all movies (Bollywood + Hollywood) = Rs. $(11000/200)$ = Rs. 55

Now, using allegation method, we get,



So, the ratio of the number of Bollywood and Hollywood movies released in theatre T = 15:5 = 3:1.

The number of Hollywood movies released in theatre $T = 1/4$ of $200 = 50$

And, the number of Hollywood movies released in theatre $S = 45$

Therefore, the required percentage = $\{(45/50) \times 100\} \% = 90\%$.

Hence, the correct answer is option E.

6. Questions

Subject A

Let the maximum marks in subjective and objective section be 1000p and 300p respectively.

So, the marks obtained in subjective section = 80% of 1000p = 800p

And, the passing marks in subjective section = 40% of 1000p = 400p

Also, the marks obtained in objective section = 75% of 300p = 225p

And, the passing marks in objective section = 30% of 300p = 90p

Now,

$$800p - 90p = 142$$

$$710p = 142$$

$$p = 0.20$$

Therefore,

The maximum marks in subjective section = 1000p = (1000 × 0.20) = 200

The maximum marks in objective section = 300p = (300 × 0.20) = 60

The marks obtained in subjective section = 800p = (800 × 0.20) = 160

The passing marks in subjective section = 400p = (400 × 0.20) = 80

The marks obtained in objective section = 225p = (225 × 0.20) = 45

The passing marks in objective section = 90p = (90 × 0.20) = 18

Subject B

Let the maximum marks in subjective and objective section be 1500q and 400q respectively.

So, the marks obtained in subjective section = 60% of 1500q = 900q

And, the passing marks in subjective section = 30% of 1500q = 450q

Also, the marks obtained in objective section = 65% of 400q = 260q

And, the passing marks in objective section = 10% of 400q = 40q

Now,

$$900q - 260q = 64$$

$$640q = 64$$

$$q = 0.10$$

Therefore,

The maximum marks in subjective section = 1500q = (1500 × 0.10) = 150

The maximum marks in objective section = 400q = (400 × 0.10) = 40

The marks obtained in subjective section = 900q = (900 × 0.10) = 90

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The passing marks in subjective section = $450q = (450 \times 0.10) = 45$

The marks obtained in objective section = $260q = (260 \times 0.10) = 26$

The passing marks in objective section = $40q = (40 \times 0.10) = 4$

Subject C

Let the maximum marks in subjective and objective section be $500r$ and $400r$ respectively.

So, the marks obtained in subjective section = 90% of $500r = 450r$

And, the passing marks in subjective section = 20% of $500r = 100r$

Also, the marks obtained in objective section = 50% of $400r = 200r$

And, the passing marks in objective section = 60% of $400r = 240r$

Now,

$$100r + 240r = (34 \times 2)$$

$$340r = 68$$

$$r = 0.20$$

Therefore,

The maximum marks in subjective section = $500r = (500 \times 0.20) = 100$

The maximum marks in objective section = $400r = (400 \times 0.20) = 80$

The marks obtained in subjective section = $450r = (450 \times 0.20) = 90$

The passing marks in subjective section = $100r = (100 \times 0.20) = 20$

The marks obtained in objective section = $200r = (200 \times 0.20) = 40$

The passing marks in objective section = $240r = (240 \times 0.20) = 48$

Subject D

Let the maximum marks in subjective and objective section be $1000s$ and $300s$ respectively.

So, the marks obtained in subjective section = 70% of $1000s = 700s$

And, the passing marks in subjective section = 50% of $1000s = 500s$

Also, the marks obtained in objective section = 60% of $300s = 180s$

And, the passing marks in objective section = 40% of $300s = 120s$

Now,

$$500s - 180s = 80$$

$$320s = 80$$

$$s = 0.25$$

Therefore,

The maximum marks in subjective section = $1000s = (1000 \times 0.25) = 250$

The maximum marks in objective section = $300s = (300 \times 0.25) = 75$

The marks obtained in subjective section = $700s = (700 \times 0.25) = 175$

The passing marks in subjective section = $500s = (500 \times 0.25) = 125$

The marks obtained in objective section = $180s = (180 \times 0.25) = 45$

The passing marks in objective section = $120s = (120 \times 0.20) = 30$

Subjective Section

Subject	The maximum marks	The marks obtained	The passing marks
A	200	160	80
B	150	90	45
C	100	90	20
D	250	175	125
Total	700	515	270

Objective Section

Subject	The maximum marks	The marks obtained	The passing marks
A	60	45	18
B	40	26	4
C	80	40	48
D	75	45	30
Total	255	156	100

Answer: D

Now, according to the question,

Subject A (subjective section)

The maximum marks = 200

The marks awarded for each correct attempt = 2

So, the total number of questions attempted = $(200/2) = 100$ (*all the contained questions are attempted*)

Let the number of correct attempts be n .

So, the number of wrong attempts = $(100-n)$

Therefore, the total marks obtained = $[(n \times 2) - \{(100-n) \times 0.50\}] = \{2n - (50 - 0.50n)\} = (2.50n - 50)$

Now,

$$2.50n - 50 = 160$$

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$$2.50n = 210$$

$$n = 84$$

So, the number of correct attempts = $n = 84$

And, number of wrong attempts = $(100-n) = (100-84) = 16$

Therefore, the required ratio = $84:16 = 21:4$.

Hence, the correct answer is option D.

7. Questions

Subject A

Let the maximum marks in subjective and objective section be $1000p$ and $300p$ respectively.

So, the marks obtained in subjective section = 80% of $1000p = 800p$

And, the passing marks in subjective section = 40% of $1000p = 400p$

Also, the marks obtained in objective section = 75% of $300p = 225p$

And, the passing marks in objective section = 30% of $300p = 90p$

Now,

$$800p - 90p = 142$$

$$710p = 142$$

$$p = 0.20$$

Therefore,

The maximum marks in subjective section = $1000p = (1000 \times 0.20) = 200$

The maximum marks in objective section = $300p = (300 \times 0.20) = 60$

The marks obtained in subjective section = $800p = (800 \times 0.20) = 160$

The passing marks in subjective section = $400p = (400 \times 0.20) = 80$

The marks obtained in objective section = $225p = (225 \times 0.20) = 45$

The passing marks in objective section = $90p = (90 \times 0.20) = 18$

Subject B

Let the maximum marks in subjective and objective section be $1500q$ and $400q$ respectively.

So, the marks obtained in subjective section = 60% of $1500q = 900q$

And, the passing marks in subjective section = 30% of $1500q = 450q$

Also, the marks obtained in objective section = 65% of $400q = 260q$

And, the passing marks in objective section = 10% of $400q = 40q$

Now,

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$$900q - 260q = 64$$

$$640q = 64$$

$$q = 0.10$$

Therefore,

$$\text{The maximum marks in subjective section} = 1500q = (1500 \times 0.10) = 150$$

$$\text{The maximum marks in objective section} = 400q = (400 \times 0.10) = 40$$

$$\text{The marks obtained in subjective section} = 900q = (900 \times 0.10) = 90$$

$$\text{The passing marks in subjective section} = 450q = (450 \times 0.10) = 45$$

$$\text{The marks obtained in objective section} = 260q = (260 \times 0.10) = 26$$

$$\text{The passing marks in objective section} = 40q = (40 \times 0.10) = 4$$

Subject C

Let the maximum marks in subjective and objective section be $500r$ and $400r$ respectively.

$$\text{So, the marks obtained in subjective section} = 90\% \text{ of } 500r = 450r$$

$$\text{And, the passing marks in subjective section} = 20\% \text{ of } 500r = 100r$$

$$\text{Also, the marks obtained in objective section} = 50\% \text{ of } 400r = 200r$$

$$\text{And, the passing marks in objective section} = 60\% \text{ of } 400r = 240r$$

Now,

$$100r + 240r = (34 \times 2)$$

$$340r = 68$$

$$r = 0.20$$

Therefore,

$$\text{The maximum marks in subjective section} = 500r = (500 \times 0.20) = 100$$

$$\text{The maximum marks in objective section} = 400r = (400 \times 0.20) = 80$$

$$\text{The marks obtained in subjective section} = 450r = (450 \times 0.20) = 90$$

$$\text{The passing marks in subjective section} = 100r = (100 \times 0.20) = 20$$

$$\text{The marks obtained in objective section} = 200r = (200 \times 0.20) = 40$$

$$\text{The passing marks in objective section} = 240r = (240 \times 0.20) = 48$$

Subject D

Let the maximum marks in subjective and objective section be $1000s$ and $300s$ respectively.

$$\text{So, the marks obtained in subjective section} = 70\% \text{ of } 1000s = 700s$$

$$\text{And, the passing marks in subjective section} = 50\% \text{ of } 1000s = 500s$$

Also, the marks obtained in objective section = 60% of 300s = 180s

And, the passing marks in objective section = 40% of 300s = 120s

Now,

$$500s - 180s = 80$$

$$320s = 80$$

$$s = 0.25$$

Therefore,

$$\text{The maximum marks in subjective section} = 1000s = (1000 \times 0.25) = 250$$

$$\text{The maximum marks in objective section} = 300s = (300 \times 0.25) = 75$$

$$\text{The marks obtained in subjective section} = 700s = (700 \times 0.25) = 175$$

$$\text{The passing marks in subjective section} = 500s = (500 \times 0.25) = 125$$

$$\text{The marks obtained in objective section} = 180s = (180 \times 0.25) = 45$$

$$\text{The passing marks in objective section} = 120s = (120 \times 0.25) = 30$$

Subjective Section

Subject	The maximum marks	The marks obtained	The passing marks
A	200	160	80
B	150	90	45
C	100	90	20
D	250	175	125
Total	700	515	270

Objective Section

Subject	The maximum marks	The marks obtained	The passing marks
A	60	45	18
B	40	26	4
C	80	40	48
D	75	45	30
Total	255	156	100

Answer: A

Subject B

Subjective Section

The total maximum marks (part I + part II) = 150

The maximum marks in part I = $\frac{2}{3}$ of 150 = 100

The maximum marks in part II = $\frac{1}{3}$ of 150 = 50

The total marks obtained (part I + part II) = 90

The marks obtained in part I = 50% of 100 = 50

So, the marks obtained in part II = $(90-50) = 40$

Objective Section

The total maximum marks (part I + part II) = 40

The maximum marks in part I = $\frac{3}{4}$ of 40 = 30

The maximum marks in part II = $\frac{1}{4}$ of 40 = 10

The total marks obtained (part I + part II) = 26

The marks obtained in part I = 60% of 30 = 18

So, the marks obtained in part II = $(26-18) = 8$

Therefore, the required ratio = $40:8 = 5:1$.

Hence, the correct answer is option A.

8. Questions

Subject A

Let the maximum marks in subjective and objective section be 1000p and 300p respectively.

So, the marks obtained in subjective section = 80% of 1000p = 800p

And, the passing marks in subjective section = 40% of 1000p = 400p

Also, the marks obtained in objective section = 75% of 300p = 225p

And, the passing marks in objective section = 30% of 300p = 90p

Now,

$$800p - 90p = 142$$

$$710p = 142$$

$$p = 0.20$$

Therefore,

The maximum marks in subjective section = $1000p = (1000 \times 0.20) = 200$

The maximum marks in objective section = $300p = (300 \times 0.20) = 60$

The marks obtained in subjective section = $800p = (800 \times 0.20) = 160$

The passing marks in subjective section = $400p = (400 \times 0.20) = 80$

The marks obtained in objective section = $225p = (225 \times 0.20) = 45$

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The passing marks in objective section = $90p = (90 \times 0.20) = 18$

Subject B

Let the maximum marks in subjective and objective section be $1500q$ and $400q$ respectively.

So, the marks obtained in subjective section = 60% of $1500q = 900q$

And, the passing marks in subjective section = 30% of $1500q = 450q$

Also, the marks obtained in objective section = 65% of $400q = 260q$

And, the passing marks in objective section = 10% of $400q = 40q$

Now,

$$900q - 260q = 64$$

$$640q = 64$$

$$q = 0.10$$

Therefore,

The maximum marks in subjective section = $1500q = (1500 \times 0.10) = 150$

The maximum marks in objective section = $400q = (400 \times 0.10) = 40$

The marks obtained in subjective section = $900q = (900 \times 0.10) = 90$

The passing marks in subjective section = $450q = (450 \times 0.10) = 45$

The marks obtained in objective section = $260q = (260 \times 0.10) = 26$

The passing marks in objective section = $40q = (40 \times 0.10) = 4$

Subject C

Let the maximum marks in subjective and objective section be $500r$ and $400r$ respectively.

So, the marks obtained in subjective section = 90% of $500r = 450r$

And, the passing marks in subjective section = 20% of $500r = 100r$

Also, the marks obtained in objective section = 50% of $400r = 200r$

And, the passing marks in objective section = 60% of $400r = 240r$

Now,

$$100r + 240r = (34 \times 2)$$

$$340r = 68$$

$$r = 0.20$$

Therefore,

The maximum marks in subjective section = $500r = (500 \times 0.20) = 100$

The maximum marks in objective section = $400r = (400 \times 0.20) = 80$

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The marks obtained in subjective section = $450r = (450 \times 0.20) = 90$

The passing marks in subjective section = $100r = (100 \times 0.20) = 20$

The marks obtained in objective section = $200r = (200 \times 0.20) = 40$

The passing marks in objective section = $240r = (240 \times 0.20) = 48$

Subject D

Let the maximum marks in subjective and objective section be 1000s and 300s respectively.

So, the marks obtained in subjective section = 70% of 1000s = 700s

And, the passing marks in subjective section = 50% of 1000s = 500s

Also, the marks obtained in objective section = 60% of 300s = 180s

And, the passing marks in objective section = 40% of 300s = 120s

Now,

$$500s - 180s = 80$$

$$320s = 80$$

$$s = 0.25$$

Therefore,

The maximum marks in subjective section = $1000s = (1000 \times 0.25) = 250$

The maximum marks in objective section = $300s = (300 \times 0.25) = 75$

The marks obtained in subjective section = $700s = (700 \times 0.25) = 175$

The passing marks in subjective section = $500s = (500 \times 0.25) = 125$

The marks obtained in objective section = $180s = (180 \times 0.25) = 45$

The passing marks in objective section = $120s = (120 \times 0.20) = 30$

Subjective Section

Subject	The maximum marks	The marks obtained	The passing marks
A	200	160	80
B	150	90	45
C	100	90	20
D	250	175	125
Total	700	515	270

Objective Section

Subject	The maximum marks	The marks obtained	The passing marks
A	60	45	18
B	40	26	4
C	80	40	48
D	75	45	30
Total	255	156	100

Answer: D

Subject E (subjective section)

The maximum marks = 300

The marks obtained = $X\%$ of 300 = $3X$

The passing marks = $(3X+30)$

Now,

$$(3X+30) - (X-20) = 3X$$

$$X - 20 = 30$$

$$X = 50$$

I: 50 is exactly divisible by both 5 and 10.

So, I is definitely true.

II: 50 is an even composite number.

So, II is definitely true.

III: The sum of the digits of 50 = $(5+0) = 5$.

5 is prime number.

So, III is definitely true.

Therefore, all I, II and III are true.

Hence, the correct answer is option D.

9. Questions

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Subject A

Let the maximum marks in subjective and objective section be 1000p and 300p respectively.

So, the marks obtained in subjective section = 80% of 1000p = 800p

And, the passing marks in subjective section = 40% of 1000p = 400p

Also, the marks obtained in objective section = 75% of 300p = 225p

And, the passing marks in objective section = 30% of 300p = 90p

Now,

$$800p - 90p = 142$$

$$710p = 142$$

$$p = 0.20$$

Therefore,

$$\text{The maximum marks in subjective section} = 1000p = (1000 \times 0.20) = 200$$

$$\text{The maximum marks in objective section} = 300p = (300 \times 0.20) = 60$$

$$\text{The marks obtained in subjective section} = 800p = (800 \times 0.20) = 160$$

$$\text{The passing marks in subjective section} = 400p = (400 \times 0.20) = 80$$

$$\text{The marks obtained in objective section} = 225p = (225 \times 0.20) = 45$$

$$\text{The passing marks in objective section} = 90p = (90 \times 0.20) = 18$$

Subject B

Let the maximum marks in subjective and objective section be $1500q$ and $400q$ respectively.

$$\text{So, the marks obtained in subjective section} = 60\% \text{ of } 1500q = 900q$$

$$\text{And, the passing marks in subjective section} = 30\% \text{ of } 1500q = 450q$$

$$\text{Also, the marks obtained in objective section} = 65\% \text{ of } 400q = 260q$$

$$\text{And, the passing marks in objective section} = 10\% \text{ of } 400q = 40q$$

Now,

$$900q - 260q = 64$$

$$640q = 64$$

$$q = 0.10$$

Therefore,

$$\text{The maximum marks in subjective section} = 1500q = (1500 \times 0.10) = 150$$

$$\text{The maximum marks in objective section} = 400q = (400 \times 0.10) = 40$$

$$\text{The marks obtained in subjective section} = 900q = (900 \times 0.10) = 90$$

$$\text{The passing marks in subjective section} = 450q = (450 \times 0.10) = 45$$

$$\text{The marks obtained in objective section} = 260q = (260 \times 0.10) = 26$$

$$\text{The passing marks in objective section} = 40q = (40 \times 0.10) = 4$$

Subject C

Let the maximum marks in subjective and objective section be $500r$ and $400r$ respectively.

$$\text{So, the marks obtained in subjective section} = 90\% \text{ of } 500r = 450r$$

And, the passing marks in subjective section = 20% of 500r = 100r

Also, the marks obtained in objective section = 50% of 400r = 200r

And, the passing marks in objective section = 60% of 400r = 240r

Now,

$$100r + 240r = (34 \times 2)$$

$$340r = 68$$

$$r = 0.20$$

Therefore,

$$\text{The maximum marks in subjective section} = 500r = (500 \times 0.20) = 100$$

$$\text{The maximum marks in objective section} = 400r = (400 \times 0.20) = 80$$

$$\text{The marks obtained in subjective section} = 450r = (450 \times 0.20) = 90$$

$$\text{The passing marks in subjective section} = 100r = (100 \times 0.20) = 20$$

$$\text{The marks obtained in objective section} = 200r = (200 \times 0.20) = 40$$

$$\text{The passing marks in objective section} = 240r = (240 \times 0.20) = 48$$

Subject D

Let the maximum marks in subjective and objective section be 1000s and 300s respectively.

$$\text{So, the marks obtained in subjective section} = 70\% \text{ of } 1000s = 700s$$

$$\text{And, the passing marks in subjective section} = 50\% \text{ of } 1000s = 500s$$

$$\text{Also, the marks obtained in objective section} = 60\% \text{ of } 300s = 180s$$

$$\text{And, the passing marks in objective section} = 40\% \text{ of } 300s = 120s$$

Now,

$$500s - 180s = 80$$

$$320s = 80$$

$$s = 0.25$$

Therefore,

$$\text{The maximum marks in subjective section} = 1000s = (1000 \times 0.25) = 250$$

$$\text{The maximum marks in objective section} = 300s = (300 \times 0.25) = 75$$

$$\text{The marks obtained in subjective section} = 700s = (700 \times 0.25) = 175$$

$$\text{The passing marks in subjective section} = 500s = (500 \times 0.25) = 125$$

$$\text{The marks obtained in objective section} = 180s = (180 \times 0.25) = 45$$

$$\text{The passing marks in objective section} = 120s = (120 \times 0.25) = 30$$

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Subjective Section

Subject	The maximum marks	The marks obtained	The passing marks
A	200	160	80
B	150	90	45
C	100	90	20
D	250	175	125
Total	700	515	270

Objective Section

Subject	The maximum marks	The marks obtained	The passing marks
A	60	45	18
B	40	26	4
C	80	40	48
D	75	45	30
Total	255	156	100

Answer: C

Subject C (objective section)

Let the number of correct and wrong attempts made be $11m$ and $4m$ respectively.

So, the total marks obtained = $\{(11m \times 1) - (4m \times 0.25)\} = (11m - m) = 10m$

Now,

$$10m = 40$$

$$m = 4$$

So, the total number of questions attempted correct = $11m = 44$

Hence, the correct answer is option C.

10. Questions

Subject A

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Let the maximum marks in subjective and objective section be 1000p and 300p respectively.

So, the marks obtained in subjective section = 80% of 1000p = 800p

And, the passing marks in subjective section = 40% of 1000p = 400p

Also, the marks obtained in objective section = 75% of 300p = 225p

And, the passing marks in objective section = 30% of 300p = 90p

Now,

$$800p - 90p = 142$$

$$710p = 142$$

$$p = 0.20$$

Therefore,

$$\text{The maximum marks in subjective section} = 1000p = (1000 \times 0.20) = 200$$

$$\text{The maximum marks in objective section} = 300p = (300 \times 0.20) = 60$$

$$\text{The marks obtained in subjective section} = 800p = (800 \times 0.20) = 160$$

$$\text{The passing marks in subjective section} = 400p = (400 \times 0.20) = 80$$

$$\text{The marks obtained in objective section} = 225p = (225 \times 0.20) = 45$$

$$\text{The passing marks in objective section} = 90p = (90 \times 0.20) = 18$$

Subject B

Let the maximum marks in subjective and objective section be $1500q$ and $400q$ respectively.

$$\text{So, the marks obtained in subjective section} = 60\% \text{ of } 1500q = 900q$$

$$\text{And, the passing marks in subjective section} = 30\% \text{ of } 1500q = 450q$$

$$\text{Also, the marks obtained in objective section} = 65\% \text{ of } 400q = 260q$$

$$\text{And, the passing marks in objective section} = 10\% \text{ of } 400q = 40q$$

Now,

$$900q - 260q = 64$$

$$640q = 64$$

$$q = 0.10$$

Therefore,

$$\text{The maximum marks in subjective section} = 1500q = (1500 \times 0.10) = 150$$

$$\text{The maximum marks in objective section} = 400q = (400 \times 0.10) = 40$$

$$\text{The marks obtained in subjective section} = 900q = (900 \times 0.10) = 90$$

$$\text{The passing marks in subjective section} = 450q = (450 \times 0.10) = 45$$

$$\text{The marks obtained in objective section} = 260q = (260 \times 0.10) = 26$$

$$\text{The passing marks in objective section} = 40q = (40 \times 0.10) = 4$$

Subject C

Let the maximum marks in subjective and objective section be $500r$ and $400r$ respectively.

$$\text{So, the marks obtained in subjective section} = 90\% \text{ of } 500r = 450r$$

$$\text{And, the passing marks in subjective section} = 20\% \text{ of } 500r = 100r$$

Also, the marks obtained in objective section = 50% of 400r = 200r

And, the passing marks in objective section = 60% of 400r = 240r

Now,

$$100r + 240r = (34 \times 2)$$

$$340r = 68$$

$$r = 0.20$$

Therefore,

$$\text{The maximum marks in subjective section} = 500r = (500 \times 0.20) = 100$$

$$\text{The maximum marks in objective section} = 400r = (400 \times 0.20) = 80$$

$$\text{The marks obtained in subjective section} = 450r = (450 \times 0.20) = 90$$

$$\text{The passing marks in subjective section} = 100r = (100 \times 0.20) = 20$$

$$\text{The marks obtained in objective section} = 200r = (200 \times 0.20) = 40$$

$$\text{The passing marks in objective section} = 240r = (240 \times 0.20) = 48$$

Subject D

Let the maximum marks in subjective and objective section be 1000s and 300s respectively.

$$\text{So, the marks obtained in subjective section} = 70\% \text{ of } 1000s = 700s$$

$$\text{And, the passing marks in subjective section} = 50\% \text{ of } 1000s = 500s$$

$$\text{Also, the marks obtained in objective section} = 60\% \text{ of } 300s = 180s$$

$$\text{And, the passing marks in objective section} = 40\% \text{ of } 300s = 120s$$

Now,

$$500s - 180s = 80$$

$$320s = 80$$

$$s = 0.25$$

Therefore,

$$\text{The maximum marks in subjective section} = 1000s = (1000 \times 0.25) = 250$$

$$\text{The maximum marks in objective section} = 300s = (300 \times 0.25) = 75$$

$$\text{The marks obtained in subjective section} = 700s = (700 \times 0.25) = 175$$

$$\text{The passing marks in subjective section} = 500s = (500 \times 0.25) = 125$$

$$\text{The marks obtained in objective section} = 180s = (180 \times 0.25) = 45$$

$$\text{The passing marks in objective section} = 120s = (120 \times 0.25) = 30$$

Subjective Section

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Subject	The maximum marks	The marks obtained	The passing marks
A	200	160	80
B	150	90	45
C	100	90	20
D	250	175	125
Total	700	515	270

Objective Section

Subject	The maximum marks	The marks obtained	The passing marks
A	60	45	18
B	40	26	4
C	80	40	48
D	75	45	30
Total	255	156	100

Answer: B

Subject D

Objective Section

The total maximum marks = 75

The maximum marks from practical = 30

The maximum marks from theory = $(75-30) = 45$

Subjective Section

The total maximum marks = 250

The maximum marks from theory = $(45+155) = 200$

The maximum marks from practical = $(250-200) = 50$

Therefore, the value of question mark = $\{(50/250) \times 100\} = 20$.

Hence, the correct answer is option B.

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11. Questions

School X

The ratio of the number of students who like only sports and only arts = $240:100 = 12:5$. Let it be $12m$ and $5m$ respectively.

The number of students who like both sports and arts = 30

So, the total number of students who like sports = $(12m+30)$

And the total number of students who like arts = $(5m+30)$

Now,

$$(12m+30)/(5m+30) = 15/8$$

$$96m + 240 = 75m + 450$$

$$21m = 210$$

$$m = 10$$

Therefore,

$$\text{The number of students who like only sports} = 12m = (12 \times 10) = 120$$

$$\text{The number of students who like only arts} = 5m = (5 \times 10) = 50$$

$$\text{The number of students who like both sports and arts} = 30$$

$$\text{The total number of students} = (120+50+30) = 200$$

School Y

Let the number of students who like only sports and only arts be $3n$ and $2n$ respectively.

$$\text{The number of students who like both sports and arts} = (30+20) = 50$$

$$\text{So, the total number of students who like sports} = (3n+50)$$

$$\text{And the total number of students who like arts} = (2n+50)$$

Now,

$$(3n+50)/(2n+50) = 4/3$$

$$9n + 150 = 8n + 200$$

$$n = 50$$

Therefore,

$$\text{The number of students who like only sports} = 3n = (3 \times 50) = 150$$

$$\text{The number of students who like only arts} = 2n = (2 \times 50) = 100$$

$$\text{The number of students who like both sports and arts} = 50$$

$$\text{The total number of students} = (150+100+50) = 300$$

School Z

Let the number of students who like only arts be k .

$$\text{So, the number of students who like only sports} = (k+85)$$

$$\text{The number of students who like both sports and arts} = 15$$

$$\text{So, the total number of students who like sports} = \{(k+85)+15\} = (k+100)$$

$$\text{And the total number of students who like arts} = (k+15)$$

Now,

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$$(k+100) + (k+15) = 265$$

$$2k + 115 = 265$$

$$2k = 150$$

$$k = 75$$

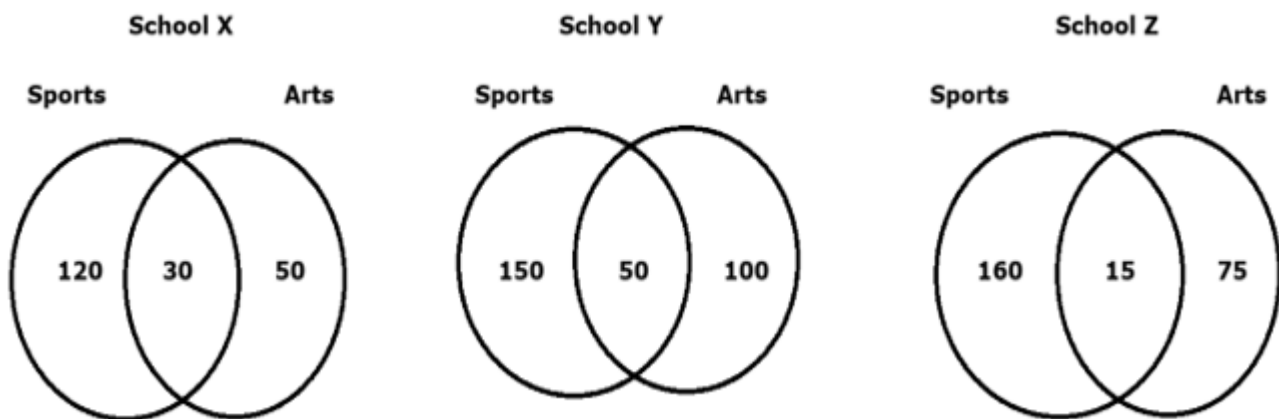
Therefore,

$$\text{The number of students who like only sports} = (k+85) = (75+85) = 160$$

$$\text{The number of students who like only arts} = k = 75$$

$$\text{The number of students who like both sports and arts} = 15$$

$$\text{The total number of students} = (160+75+15) = 250$$



Answer: B

Now, according to the question,

School X

$$\text{The total number of students who like sports} = (120+30) = 150$$

$$\text{The total number of students who like arts} = (50+30) = 80$$

Now, let the total number of students who like cricket and dancing be $25a$ and $7a$ respectively.

$$\text{So, the total number of students who like football} = (150-25a)$$

$$\text{And, total number of students who like singing} = (80-7a)$$

Now,

$$(80-7a) - (150-25a) = 2$$

$$78 - 7a = 150 - 25a$$

$$18a = 72$$

$$a = 4$$

$$\text{So, total number of students who like cricket} = 25a = (25 \times 4) = 100$$

$$\text{And, total number of students who like singing} = (80-7a) = (80-7 \times 4) = (80-28) = 52$$

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Therefore, the required difference = $(100-52) = 48$.

Hence, the correct answer is option B.

12. Questions

School X

The ratio of the number of students who like only sports and only arts = $240:100 = 12:5$. Let it be $12m$ and $5m$ respectively.

The number of students who like both sports and arts = 30

So, the total number of students who like sports = $(12m+30)$

And the total number of students who like arts = $(5m+30)$

Now,

$$(12m+30)/(5m+30) = 15/8$$

$$96m + 240 = 75m + 450$$

$$21m = 210$$

$$m = 10$$

Therefore,

The number of students who like only sports = $12m = (12 \times 10) = 120$

The number of students who like only arts = $5m = (5 \times 10) = 50$

The number of students who like both sports and arts = 30

The total number of students = $(120+50+30) = 200$

School Y

Let the number of students who like only sports and only arts be $3n$ and $2n$ respectively.

The number of students who like both sports and arts = $(30+20) = 50$

So, the total number of students who like sports = $(3n+50)$

And the total number of students who like arts = $(2n+50)$

Now,

$$(3n+50)/(2n+50) = 4/3$$

$$9n + 150 = 8n + 200$$

$$n = 50$$

Therefore,

The number of students who like only sports = $3n = (3 \times 50) = 150$

The number of students who like only arts = $2n = (2 \times 50) = 100$

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The number of students who like both sports and arts = 50

The total number of students = $(150+100+50) = 300$

School Z

Let the number of students who like only arts be k .

So, the number of students who like only sports = $(k+85)$

The number of students who like both sports and arts = 15

So, the total number of students who like sports = $\{(k+85)+15\} = (k+100)$

And the total number of students who like arts = $(k+15)$

Now,

$$(k+100) + (k+15) = 265$$

$$2k + 115 = 265$$

$$2k = 150$$

$$k = 75$$

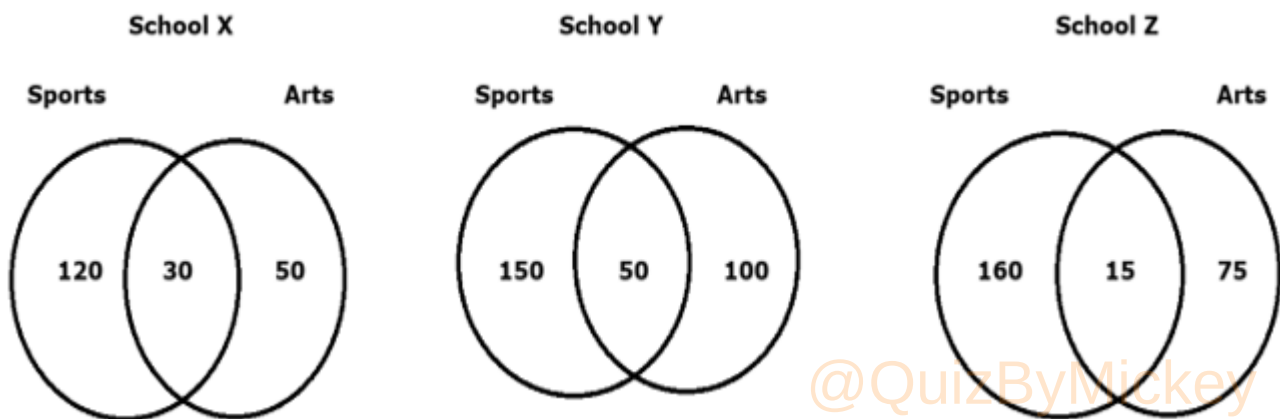
Therefore,

The number of students who like only sports = $(k+85) = (75+85) = 160$

The number of students who like only arts = $k = 75$

The number of students who like both sports and arts = 15

The total number of students = $(160+75+15) = 250$



Answer: C

School Y

The total number of students who like sports = $(150+50) = 200$

The total number of students who like arts = $(100+50) = 150$

Using I

The total number of boys who like sports = 60% of $200 = 120$

The total number of girls who like arts = 20% of 150 = 30

The total number of girls who like sports = (200-120) = 80

The total number of boys who like arts = (150-30) = 120

So, the required ratio = 80:120 = 2:3.

Therefore, I does not satisfy.

Using II

The total number of boys who like sports = 80% of 200 = 160

The total number of girls who like arts = 30% of 150 = 45

The total number of girls who like sports = (200-160) = 40

The total number of boys who like arts = (150-45) = 105

So, the required ratio = 40:105 = 8:21.

Therefore, II does satisfy.

Using III

The total number of boys who like sports = 75% of 200 = 150

The total number of girls who like arts = 10% of 150 = 15

The total number of girls who like sports = (200-150) = 50

The total number of boys who like arts = (150-15) = 135

So, the required ratio = 50:135 = 10:27.

Therefore, III does not satisfy.

So, only II follow.

Hence, the correct answer is option C.

13. Questions

School X

The ratio of the number of students who like only sports and only arts = 240:100 = 12:5. Let it be 12m and 5m respectively.

The number of students who like both sports and arts = 30

So, the total number of students who like sports = (12m+30)

And the total number of students who like arts = (5m+30)

Now,

$$(12m+30)/(5m+30) = 15/8$$

$$96m + 240 = 75m + 450$$

$$21m = 210$$

$$m = 10$$

Therefore,

$$\text{The number of students who like only sports} = 12m = (12 \times 10) = 120$$

$$\text{The number of students who like only arts} = 5m = (5 \times 10) = 50$$

$$\text{The number of students who like both sports and arts} = 30$$

$$\text{The total number of students} = (120 + 50 + 30) = 200$$

School Y

Let the number of students who like only sports and only arts be $3n$ and $2n$ respectively.

$$\text{The number of students who like both sports and arts} = (30 + 20) = 50$$

$$\text{So, the total number of students who like sports} = (3n + 50)$$

$$\text{And the total number of students who like arts} = (2n + 50)$$

Now,

$$(3n + 50) / (2n + 50) = 4/3$$

$$9n + 150 = 8n + 200$$

$$n = 50$$

Therefore,

$$\text{The number of students who like only sports} = 3n = (3 \times 50) = 150$$

$$\text{The number of students who like only arts} = 2n = (2 \times 50) = 100$$

$$\text{The number of students who like both sports and arts} = 50$$

$$\text{The total number of students} = (150 + 100 + 50) = 300$$

School Z

Let the number of students who like only arts be k .

$$\text{So, the number of students who like only sports} = (k + 85)$$

$$\text{The number of students who like both sports and arts} = 15$$

$$\text{So, the total number of students who like sports} = \{(k + 85) + 15\} = (k + 100)$$

$$\text{And the total number of students who like arts} = (k + 15)$$

Now,

$$(k + 100) + (k + 15) = 265$$

$$2k + 115 = 265$$

$$2k = 150$$

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$$k = 75$$

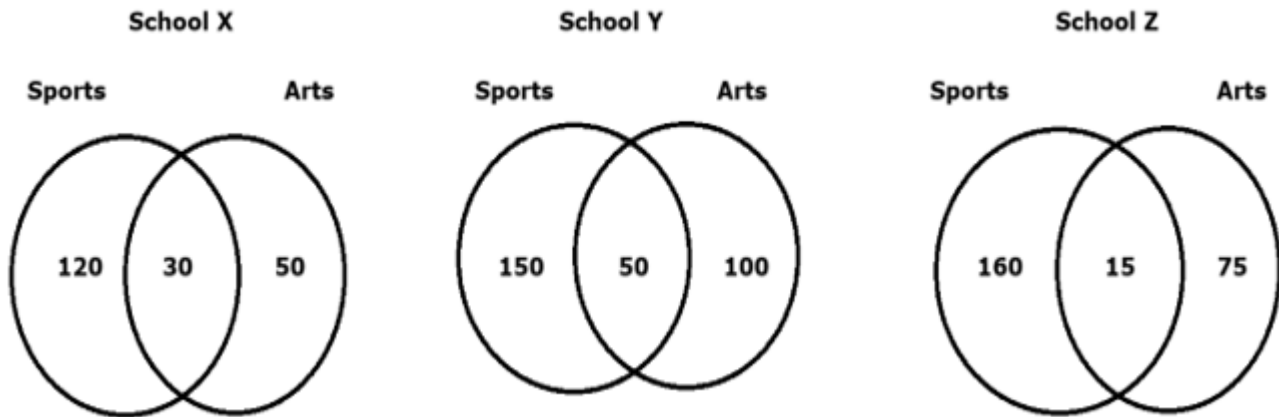
Therefore,

$$\text{The number of students who like only sports} = (k+85) = (75+85) = 160$$

$$\text{The number of students who like only arts} = k = 75$$

$$\text{The number of students who like both sports and arts} = 15$$

$$\text{The total number of students} = (160+75+15) = 250$$



Answer: A

School Z

$$\text{The total number of students who likes sports} = 175$$

$$\text{The total number of scholarship holder students who likes sports} = (2P+5)$$

$$\text{So, the total number of non-scholarship holder students who likes sports} = \{6 \times (2P+5)\} = (12P+30)$$

Now,

$$(2P+5) + (12P+30) = 175$$

$$14P + 35 = 175$$

$$14P = 140$$

$$P = 10$$

Therefore, question mark = 10.

Hence, the correct answer is option A.

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14. Questions

School X

The ratio of the number of students who like only sports and only arts = 240:100 = 12:5. Let it be 12m and 5m respectively.

$$\text{The number of students who like both sports and arts} = 30$$

$$\text{So, the total number of students who like sports} = (12m+30)$$

And the total number of students who like arts = $(5m+30)$

Now,

$$(12m+30)/(5m+30) = 15/8$$

$$96m + 240 = 75m + 450$$

$$21m = 210$$

$$m = 10$$

Therefore,

$$\text{The number of students who like only sports} = 12m = (12 \times 10) = 120$$

$$\text{The number of students who like only arts} = 5m = (5 \times 10) = 50$$

$$\text{The number of students who like both sports and arts} = 30$$

$$\text{The total number of students} = (120+50+30) = 200$$

School Y

Let the number of students who like only sports and only arts be $3n$ and $2n$ respectively.

$$\text{The number of students who like both sports and arts} = (30+20) = 50$$

$$\text{So, the total number of students who like sports} = (3n+50)$$

$$\text{And the total number of students who like arts} = (2n+50)$$

Now,

$$(3n+50)/(2n+50) = 4/3$$

$$9n + 150 = 8n + 200$$

$$n = 50$$

Therefore,

$$\text{The number of students who like only sports} = 3n = (3 \times 50) = 150$$

$$\text{The number of students who like only arts} = 2n = (2 \times 50) = 100$$

$$\text{The number of students who like both sports and arts} = 50$$

$$\text{The total number of students} = (150+100+50) = 300$$

School Z

Let the number of students who like only arts be k .

$$\text{So, the number of students who like only sports} = (k+85)$$

$$\text{The number of students who like both sports and arts} = 15$$

$$\text{So, the total number of students who like sports} = \{(k+85)+15\} = (k+100)$$

$$\text{And the total number of students who like arts} = (k+15)$$

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Now,

$$(k+100) + (k+15) = 265$$

$$2k + 115 = 265$$

$$2k = 150$$

$$k = 75$$

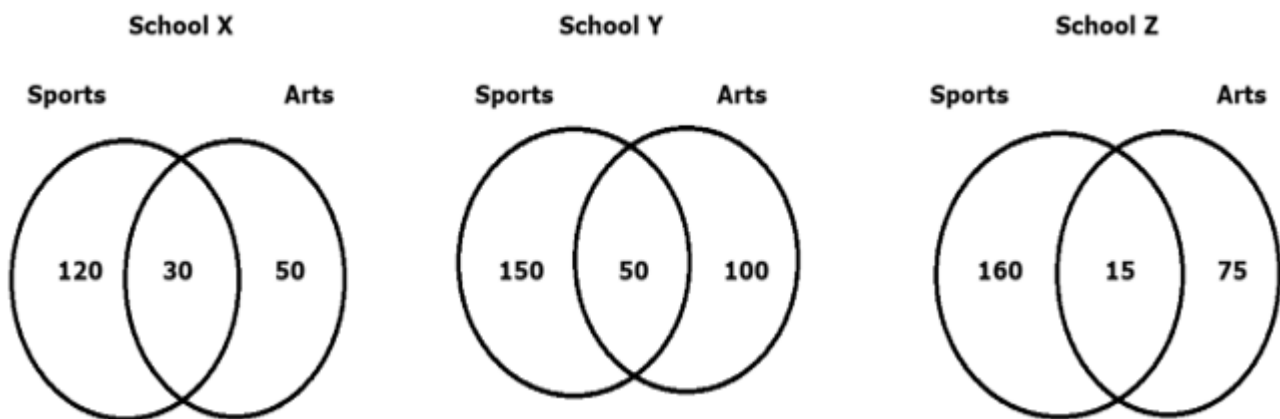
Therefore,

$$\text{The number of students who like only sports} = (k+85) = (75+85) = 160$$

$$\text{The number of students who like only arts} = k = 75$$

$$\text{The number of students who like both sports and arts} = 15$$

$$\text{The total number of students} = (160+75+15) = 250$$



Answer: D

School W

The ratio of the number of students who like only sports and only arts = 150:100 = 3:2. Let it be 3t and 2t respectively.

$$\text{The number of students who like both sports and arts} = 100$$

$$\text{So, the total number of students who like sports} = (3t+100)$$

$$\text{And, the total number of students who like arts} = (2t+100)$$

Now,

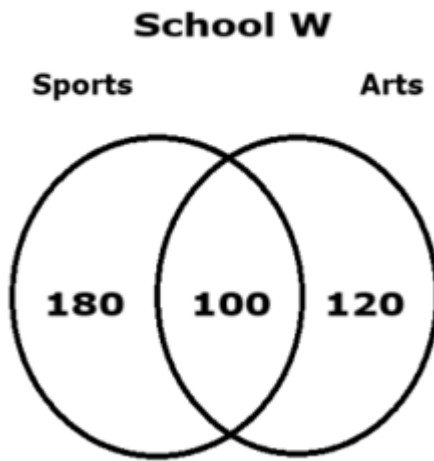
$$(3t+100)/(2t+100) = 14/11$$

$$33t + 1100 = 28t + 1400$$

$$5t = 300$$

$$t = 60$$

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So, the total number of students in school W = $(3t+2t+100) = (5t+100) = (5 \times 60+100) = (300+100) = 400$.

And, the total number of students in school X = 200

Therefore, the required percentage = $\{(200/400) \times 100\} \% = 50\%$.

Hence, the correct answer is option D.

15. Questions

School X

The ratio of the number of students who like only sports and only arts = $240:100 = 12:5$. Let it be $12m$ and $5m$ respectively.

The number of students who like both sports and arts = 30

So, the total number of students who like sports = $(12m+30)$

And the total number of students who like arts = $(5m+30)$

Now,

$$(12m+30)/(5m+30) = 15/8$$

$$96m + 240 = 75m + 450$$

$$21m = 210$$

$$m = 10$$

Therefore,

The number of students who like only sports = $12m = (12 \times 10) = 120$

The number of students who like only arts = $5m = (5 \times 10) = 50$

The number of students who like both sports and arts = 30

The total number of students = $(120+50+30) = 200$

School Y

Let the number of students who like only sports and only arts be $3n$ and $2n$ respectively.

The number of students who like both sports and arts = $(30+20) = 50$

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So, the total number of students who like sports = $(3n+50)$

And the total number of students who like arts = $(2n+50)$

Now,

$$(3n+50)/(2n+50) = 4/3$$

$$9n + 150 = 8n + 200$$

$$n = 50$$

Therefore,

The number of students who like only sports = $3n = (3 \times 50) = 150$

The number of students who like only arts = $2n = (2 \times 50) = 100$

The number of students who like both sports and arts = 50

The total number of students = $(150+100+50) = 300$

School Z

Let the number of students who like only arts be k .

So, the number of students who like only sports = $(k+85)$

The number of students who like both sports and arts = 15

So, the total number of students who like sports = $\{(k+85)+15\} = (k+100)$

And the total number of students who like arts = $(k+15)$

Now,

$$(k+100) + (k+15) = 265$$

$$2k + 115 = 265$$

$$2k = 150$$

$$k = 75$$

Therefore,

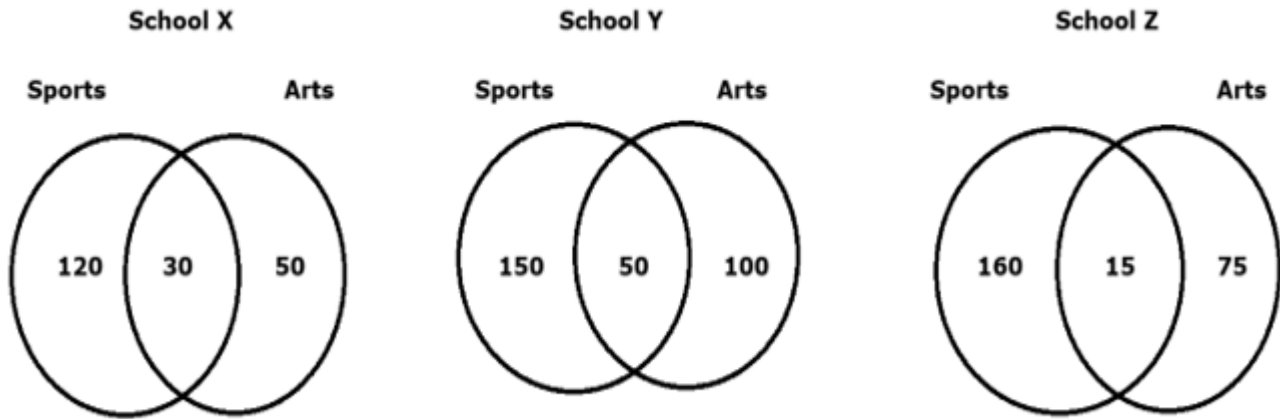
The number of students who like only sports = $(k+85) = (75+85) = 160$

The number of students who like only arts = $k = 75$

The number of students who like both sports and arts = 15

The total number of students = $(160+75+15) = 250$

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Answer: E

School X

The total number of students = 200

Let the number of boys be t .

So, the number of girls = $(t-120)$

Now,

$$t + (t-120) = 200$$

$$2t = 320$$

$$t = 160$$

So, the number of boys = $t = 160$

And, the number of girls = $(t-120) = (160-120) = 40$.

Now, the ratio of the number of boys and girls = $160:40 = 4:1$

So, $Q:R = 4:1$

$$Q = 4$$

$$R = 1$$

Therefore, the required value = $(10Q\% \text{ of } 50R) = \{(10 \times 4)\% \text{ of } (50 \times 1)\} = (40\% \text{ of } 50) = 20$.

Hence, the correct answer is option E.

16. Questions

The ratio of the total number of orders placed Tuesday and Thursday = 1:2. Let it be $100m$ and $200m$ respectively.

Tuesday

The total number of orders placed = $100m$

The number of non-delivered orders = $20\% \text{ of } 100m = 20m$

The number of delivered orders = $(100m-20m) = 80m$

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The number of delivered orders accepted = $\frac{5}{8}$ of $80m = 50m$

The number of delivered orders returned = $\frac{3}{8}$ of $80m = 30m$

Thursday

The total number of orders placed = $200m$

The number of non-delivered orders = 15% of $200m = 30m$

The number of delivered orders = $(200m - 30m) = 170m$

The number of delivered orders accepted = $\frac{11}{17}$ of $170m = 110m$

The number of delivered orders returned = $\frac{6}{17}$ of $170m = 60m$

Now,

$$60m - 50m = 20$$

$$10m = 20$$

$$m = 2$$

Also,

Let the ratio of the total number of orders placed Monday and Friday = $120:100 = 6:5$. Let it be $600n$ and $500n$ respectively.

Monday

The total number of orders placed = $600n$

The number of non-delivered orders = 25% of $600n = 150n$

The number of delivered orders = $(600n - 150n) = 450n$

The number of delivered orders accepted = $\frac{5}{6}$ of $450n = 375n$

The number of delivered orders returned = $\frac{1}{6}$ of $450n = 75n$

Friday

The total number of orders placed = $500n$

The number of non-delivered orders = 30% of $500n = 150n$

The number of delivered orders = $(500n - 150n) = 350n$

The number of delivered orders accepted = $\frac{7}{10}$ of $350n = 245n$

The number of delivered orders returned = $\frac{3}{10}$ of $350n = 105n$

Now,

$$245n - 105n = 140n$$

$$140n = 102$$

$$n = 0.60$$

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Also,

Let the total number of orders placed on Wednesday be 100k

So, the total number of non-delivered orders on Wednesday = 10% of 100k = 10k

The number of delivered orders on Wednesday = (100k-10k) = 90k

The number of delivered orders accepted on Wednesday = $\frac{3}{5}$ of 90k = 54k

The number of delivered orders returned on Wednesday = $\frac{2}{5}$ of 90k = 36k

Now,

$$36k - 10k = 39$$

$$26k = 39$$

$$k = 1.50$$

Monday

The total number of orders placed = $600n = (600 \times 0.60) = 360$

The number of non-delivered orders = $150n = (150 \times 0.60) = 90$

The number of delivered orders = $450n = (450 \times 0.60) = 270$

The number of delivered orders accepted = $375n = (375 \times 0.60) = 225$

The number of delivered orders returned = $75n = (75 \times 0.60) = 45$

Tuesday

The total number of orders placed = $100m = (100 \times 2) = 200$

The number of non-delivered orders = $20m = (20 \times 2) = 40$

The number of delivered orders = $80m = (80 \times 2) = 160$

The number of delivered orders accepted = $50m = (50 \times 2) = 100$

The number of delivered orders returned = $30m = (30 \times 2) = 60$

Wednesday

The total number of orders placed = $100k = (100 \times 1.50) = 150$

The total number of non-delivered orders = $10k = (10 \times 1.50) = 15$

The number of delivered orders = $90k = (90 \times 1.50) = 135$

The number of delivered orders accepted = $54k = (54 \times 1.50) = 81$

The number of delivered orders returned = $36k = (36 \times 1.50) = 54$

Thursday

The total number of orders placed = $200m = (200 \times 2) = 400$

The number of non-delivered orders = $30m = (30 \times 2) = 60$

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The number of delivered orders = $170m = (170 \times 2) = 340$

The number of delivered orders accepted = $110m = (110 \times 2) = 220$

The number of delivered orders returned = $60m = (60 \times 2) = 120$

Friday

The total number of orders placed = $500n = (500 \times 0.60) = 300$

The number of non-delivered orders = $150n = (150 \times 0.60) = 90$

The number of delivered orders = $350n = (350 \times 0.60) = 210$

The number of delivered orders accepted = $245n = (245 \times 0.60) = 147$

The number of delivered orders returned = $105n = (105 \times 0.60) = 63$

Days	The total number of orders placed	The number of orders delivered	The number of non-delivered orders	The number of delivered orders accepted	The number of delivered orders returned
Monday	360	270	90	225	45
Tuesday	200	160	40	100	60
Wednesday	150	135	15	81	54
Thursday	400	340	60	220	120
Friday	300	210	90	147	63
Total	1410	1115	295	773	342

Answer: B

Now, according to the question,

Monday

The total number of orders available = $(40X^2 - 40)$

The number of orders placed = 360

The number of non-placed orders = $60X$

Now,

$$360 + 60X = 40X^2 - 40$$

$$40X^2 - 60X - 400 = 0$$

$$2X^2 - 3X - 20 = 0$$

$$2X^2 - 8X + 5X - 20 = 0$$

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$$2X(X-4) + 5(X-4) = 0$$

$$(2X+5)(X-4) = 0$$

$$X = -2.50 \text{ or } +4$$

So, $X = 4$ (positive integer)

Now, the difference between the number of placed and non-placed orders on Monday = $(360-60X) = (360-60 \times 4) = (360-240) = 120$.

Therefore, the question mark = 120.

Hence, the correct answer is option B.

17. Questions

The ratio of the total number of orders placed Tuesday and Thursday = 1:2. Let it be 100m and 200m respectively.

Tuesday

The total number of orders placed = 100m

The number of non-delivered orders = 20% of 100m = 20m

The number of delivered orders = $(100m-20m) = 80m$

The number of delivered orders accepted = $\frac{5}{8}$ of 80m = 50m

The number of delivered orders returned = $\frac{3}{8}$ of 80m = 30m

Thursday

The total number of orders placed = 200m

The number of non-delivered orders = 15% of 200m = 30m

The number of delivered orders = $(200m-30m) = 170m$

The number of delivered orders accepted = $\frac{11}{17}$ of 170m = 110m

The number of delivered orders returned = $\frac{6}{17}$ of 170m = 60m

Now,

$$60m - 50m = 20$$

$$10m = 20$$

$$m = 2$$

Also,

Let the ratio of the total number of orders placed Monday and Friday = 120:100 = 6:5. Let it be 600n and 500n respectively.

Monday

The total number of orders placed = 600n

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The number of non-delivered orders = 25% of 600n = 150n

The number of delivered orders = (600n-150n) = 450n

The number of delivered orders accepted = 5/6 of 450n = 375n

The number of delivered orders returned = 1/6 of 450n = 75n

Friday

The total number of orders placed = 500n

The number of non-delivered orders = 30% of 500n = 150n

The number of delivered orders = (500n-150n) = 350n

The number of delivered orders accepted = 7/10 of 350n = 245n

The number of delivered orders returned = 3/10 of 350n = 105n

Now,

$$245n - 105n = 102$$

$$140n = 102$$

$$n = 0.60$$

Also,

Let the total number of orders placed on Wednesday be 100k

So, the total number of non-delivered orders on Wednesday = 10% of 100k = 10k

The number of delivered orders on Wednesday = (100k-10k) = 90k

The number of delivered orders accepted on Wednesday = 3/5 of 90k = 54k

The number of delivered orders returned on Wednesday = 2/5 of 90k = 36k

Now,

$$54k - 36k = 10k$$

$$18k = 10k$$

$$k = 1.50$$

Monday

The total number of orders placed = 600n = (600×0.60) = 360

The number of non-delivered orders = 150n = (150×0.60) = 90

The number of delivered orders = 450n = (450×0.60) = 270

The number of delivered orders accepted = 375n = (375×0.60) = 225

The number of delivered orders returned = 75n = (75×0.60) = 45

Tuesday

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The total number of orders placed = $100m = (100 \times 2) = 200$

The number of non-delivered orders = $20m = (20 \times 2) = 40$

The number of delivered orders = $80m = (80 \times 2) = 160$

The number of delivered orders accepted = $50m = (50 \times 2) = 100$

The number of delivered orders returned = $30m = (30 \times 2) = 60$

Wednesday

The total number of orders placed = $100k = (100 \times 1.50) = 150$

The total number of non-delivered orders = $10k = (10 \times 1.50) = 15$

The number of delivered orders = $90k = (90 \times 1.50) = 135$

The number of delivered orders accepted = $54k = (54 \times 1.50) = 81$

The number of delivered orders returned = $36k = (36 \times 1.50) = 54$

Thursday

The total number of orders placed = $200m = (200 \times 2) = 400$

The number of non-delivered orders = $30m = (30 \times 2) = 60$

The number of delivered orders = $170m = (170 \times 2) = 340$

The number of delivered orders accepted = $110m = (110 \times 2) = 220$

The number of delivered orders returned = $60m = (60 \times 2) = 120$

Friday

The total number of orders placed = $500n = (500 \times 0.60) = 300$

The number of non-delivered orders = $150n = (150 \times 0.60) = 90$

The number of delivered orders = $350n = (350 \times 0.60) = 210$

The number of delivered orders accepted = $245n = (245 \times 0.60) = 147$

The number of delivered orders returned = $105n = (105 \times 0.60) = 63$

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Days	The total number of orders placed	The number of orders delivered	The number of non-delivered orders	The number of delivered orders accepted	The number of delivered orders returned
Monday	360	270	90	225	45
Tuesday	200	160	40	100	60
Wednesday	150	135	15	81	54
Thursday	400	340	60	220	120
Friday	300	210	90	147	63
Total	1410	1115	295	773	342

Answer: A

Tuesday

The total number of orders accepted (electronic + non-electronic) = 100

The total number of orders returned (electronic + non-electronic) = 60

The ratio of the number of electronic item orders accepted and non-electronic item orders returned = 8:1.
 Let it be 8a and a respectively.

Also, the ratio of the number of non-electronic item orders accepted and electronic item orders returned = 40:100 = 2:5. Let it be 2b and 5b respectively.

Now,

$$8a + 2b = 100$$

$$4a + b = 50 \dots\dots\dots(i)$$

Also,

$$a + 5b = 60 \dots\dots\dots(ii)$$

By solving equation (i) and (ii), we get,

$$4a + b = 50$$

$$4a + 20b = 240$$

$$19b = 190$$

$$b = 10$$

Now, put the value of b in equation (i), we get,

$$4a + b = 50$$

$$4a + 10 = 50$$

$$4a = 40$$

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$$a = 10$$

Therefore, the total number of electronic items delivered (accepted + returned) on Monday = $(8a+5b) = (8 \times 10 + 5 \times 10) = (80+50) = 130$.

Hence, the correct answer is option A.

18. Questions

The ratio of the total number of orders placed Tuesday and Thursday = 1:2. Let it be 100m and 200m respectively.

Tuesday

The total number of orders placed = 100m

The number of non-delivered orders = 20% of 100m = 20m

The number of delivered orders = $(100m - 20m) = 80m$

The number of delivered orders accepted = $\frac{5}{8}$ of 80m = 50m

The number of delivered orders returned = $\frac{3}{8}$ of 80m = 30m

Thursday

The total number of orders placed = 200m

The number of non-delivered orders = 15% of 200m = 30m

The number of delivered orders = $(200m - 30m) = 170m$

The number of delivered orders accepted = $\frac{11}{17}$ of 170m = 110m

The number of delivered orders returned = $\frac{6}{17}$ of 170m = 60m

Now,

$$60m - 50m = 20$$

$$10m = 20$$

$$m = 2$$

Also,

Let the ratio of the total number of orders placed Monday and Friday = 120:100 = 6:5. Let it be 600n and 500n respectively.

Monday

The total number of orders placed = 600n

The number of non-delivered orders = 25% of 600n = 150n

The number of delivered orders = $(600n - 150n) = 450n$

The number of delivered orders accepted = $\frac{5}{6}$ of 450n = 375n

The number of delivered orders returned = $\frac{1}{6}$ of 450n = 75n

Friday

The total number of orders placed = $500n$

The number of non-delivered orders = 30% of $500n = 150n$

The number of delivered orders = $(500n - 150n) = 350n$

The number of delivered orders accepted = $7/10$ of $350n = 245n$

The number of delivered orders returned = $3/10$ of $350n = 105n$

Now,

$$245n - 75n = 102$$

$$170n = 102$$

$$n = 0.60$$

Also,

Let the total number of orders placed on Wednesday be $100k$

So, the total number of non-delivered orders on Wednesday = 10% of $100k = 10k$

The number of delivered orders on Wednesday = $(100k - 10k) = 90k$

The number of delivered orders accepted on Wednesday = $3/5$ of $90k = 54k$

The number of delivered orders returned on Wednesday = $2/5$ of $90k = 36k$

Now,

$$36k - 10k = 39$$

$$26k = 39$$

$$k = 1.50$$

Monday

The total number of orders placed = $600n = (600 \times 0.60) = 360$

The number of non-delivered orders = $150n = (150 \times 0.60) = 90$

The number of delivered orders = $450n = (450 \times 0.60) = 270$

The number of delivered orders accepted = $375n = (375 \times 0.60) = 225$

The number of delivered orders returned = $75n = (75 \times 0.60) = 45$

Tuesday

The total number of orders placed = $100m = (100 \times 2) = 200$

The number of non-delivered orders = $20m = (20 \times 2) = 40$

The number of delivered orders = $80m = (80 \times 2) = 160$

The number of delivered orders accepted = $50m = (50 \times 2) = 100$

The number of delivered orders returned = $30m = (30 \times 2) = 60$

Wednesday

The total number of orders placed = $100k = (100 \times 1.50) = 150$

The total number of non-delivered orders = $10k = (10 \times 1.50) = 15$

The number of delivered orders = $90k = (90 \times 1.50) = 135$

The number of delivered orders accepted = $54k = (54 \times 1.50) = 81$

The number of delivered orders returned = $36k = (36 \times 1.50) = 54$

Thursday

The total number of orders placed = $200m = (200 \times 2) = 400$

The number of non-delivered orders = $30m = (30 \times 2) = 60$

The number of delivered orders = $170m = (170 \times 2) = 340$

The number of delivered orders accepted = $110m = (110 \times 2) = 220$

The number of delivered orders returned = $60m = (60 \times 2) = 120$

Friday

The total number of orders placed = $500n = (500 \times 0.60) = 300$

The number of non-delivered orders = $150n = (150 \times 0.60) = 90$

The number of delivered orders = $350n = (350 \times 0.60) = 210$

The number of delivered orders accepted = $245n = (245 \times 0.60) = 147$

The number of delivered orders returned = $105n = (105 \times 0.60) = 63$

Days	The total number of orders placed	The number of orders delivered	The number of non-delivered orders	The number of delivered orders accepted	The number of delivered orders returned
Monday	360	270	90	225	45
Tuesday	200	160	40	100	60
Wednesday	150	135	15	81	54
Thursday	400	340	60	220	120
Friday	300	210	90	147	63
Total	1410	1115	295	773	342

Answer: B

Wednesday

The total number of orders accepted after the number of defective orders are returned = $(81 - Y)$

The total number of orders returned after the number of defective orders are returned = $(54+Y)$

Now,

$$(81-Y)/(54+Y) = 8/7$$

$$567 - 7Y = 432 + 8Y$$

$$15Y = 135$$

$$Y = 9$$

I: 9 is perfect square of 3.

$$3^2 = 9$$

So, I is definitely true.

II: 9 is an odd composite number.

So, II is definitely true.

III: 9 is not two-digit natural number.

So, III is definitely not true.

Therefore, both I and II are definitely true.

Hence, the correct answer is option B.

19. Questions

The ratio of the total number of orders placed Tuesday and Thursday = 1:2. Let it be 100m and 200m respectively.

Tuesday

The total number of orders placed = 100m

The number of non-delivered orders = 20% of 100m = 20m

The number of delivered orders = $(100m - 20m) = 80m$

The number of delivered orders accepted = $5/8$ of 80m = 50m

The number of delivered orders returned = $3/8$ of 80m = 30m

Thursday

The total number of orders placed = 200m

The number of non-delivered orders = 15% of 200m = 30m

The number of delivered orders = $(200m - 30m) = 170m$

The number of delivered orders accepted = $11/17$ of 170m = 110m

The number of delivered orders returned = $6/17$ of 170m = 60m

Now,

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$$60m - 50m = 20$$

$$10m = 20$$

$$m = 2$$

Also,

Let the ratio of the total number of orders placed Monday and Friday = $120:100 = 6:5$. Let it be $600n$ and $500n$ respectively.

Monday

The total number of orders placed = $600n$

The number of non-delivered orders = 25% of $600n = 150n$

The number of delivered orders = $(600n - 150n) = 450n$

The number of delivered orders accepted = $5/6$ of $450n = 375n$

The number of delivered orders returned = $1/6$ of $450n = 75n$

Friday

The total number of orders placed = $500n$

The number of non-delivered orders = 30% of $500n = 150n$

The number of delivered orders = $(500n - 150n) = 350n$

The number of delivered orders accepted = $7/10$ of $350n = 245n$

The number of delivered orders returned = $3/10$ of $350n = 105n$

Now,

$$245n - 75n = 102$$

$$170n = 102$$

$$n = 0.60$$

Also,

Let the total number of orders placed on Wednesday be $100k$

So, the total number of non-delivered orders on Wednesday = 10% of $100k = 10k$

The number of delivered orders on Wednesday = $(100k - 10k) = 90k$

The number of delivered orders accepted on Wednesday = $3/5$ of $90k = 54k$

The number of delivered orders returned on Wednesday = $2/5$ of $90k = 36k$

Now,

$$36k - 10k = 39$$

$$26k = 39$$

$k = 1.50$

Monday

The total number of orders placed = $600n = (600 \times 0.60) = 360$

The number of non-delivered orders = $150n = (150 \times 0.60) = 90$

The number of delivered orders = $450n = (450 \times 0.60) = 270$

The number of delivered orders accepted = $375n = (375 \times 0.60) = 225$

The number of delivered orders returned = $75n = (75 \times 0.60) = 45$

Tuesday

The total number of orders placed = $100m = (100 \times 2) = 200$

The number of non-delivered orders = $20m = (20 \times 2) = 40$

The number of delivered orders = $80m = (80 \times 2) = 160$

The number of delivered orders accepted = $50m = (50 \times 2) = 100$

The number of delivered orders returned = $30m = (30 \times 2) = 60$

Wednesday

The total number of orders placed = $100k = (100 \times 1.50) = 150$

The total number of non-delivered orders = $10k = (10 \times 1.50) = 15$

The number of delivered orders = $90k = (90 \times 1.50) = 135$

The number of delivered orders accepted = $54k = (54 \times 1.50) = 81$

The number of delivered orders returned = $36k = (36 \times 1.50) = 54$

Thursday

The total number of orders placed = $200m = (200 \times 2) = 400$

The number of non-delivered orders = $30m = (30 \times 2) = 60$

The number of delivered orders = $170m = (170 \times 2) = 340$

The number of delivered orders accepted = $110m = (110 \times 2) = 220$

The number of delivered orders returned = $60m = (60 \times 2) = 120$

Friday

The total number of orders placed = $500n = (500 \times 0.60) = 300$

The number of non-delivered orders = $150n = (150 \times 0.60) = 90$

The number of delivered orders = $350n = (350 \times 0.60) = 210$

The number of delivered orders accepted = $245n = (245 \times 0.60) = 147$

The number of delivered orders returned = $105n = (105 \times 0.60) = 63$

Days	The total number of orders placed	The number of orders delivered	The number of non-delivered orders	The number of delivered orders accepted	The number of delivered orders returned
Monday	360	270	90	225	45
Tuesday	200	160	40	100	60
Wednesday	150	135	15	81	54
Thursday	400	340	60	220	120
Friday	300	210	90	147	63
Total	1410	1115	295	773	342

Answer: D

The total number of orders delivered (males + females) on Thursday = 340

The total number of orders delivered (males + females) on Friday = 210

Let the number of delivered to males on Thursday be p.

So, the number of orders delivered to males on Friday = (p-90)

The number of orders delivered to females on Thursday = (340-p)

And, the number of orders delivered to females on Friday = {210-(p-90)} = (300-p)

Now,

$$(340-p)/(300-p) = 7/5$$

$$1700 - 5p = 2100 - 7p$$

$$2p = 400$$

$$p = 200$$

So, the number of orders delivered to males on Thursday = p = 200

And, the number of orders delivered to females on Thursday = (340-p) = (340-200) = 140

Therefore, the required difference = (200-140) = 60.

Hence, the correct answer is option D.

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20. Questions

The ratio of the total number of orders placed Tuesday and Thursday = 1:2. Let it be 100m and 200m respectively.

Tuesday

The total number of orders placed = 100m

The number of non-delivered orders = 20% of 100m = 20m

The number of delivered orders = $(100m - 20m) = 80m$

The number of delivered orders accepted = $5/8$ of $80m = 50m$

The number of delivered orders returned = $3/8$ of $80m = 30m$

Thursday

The total number of orders placed = $200m$

The number of non-delivered orders = 15% of $200m = 30m$

The number of delivered orders = $(200m - 30m) = 170m$

The number of delivered orders accepted = $11/17$ of $170m = 110m$

The number of delivered orders returned = $6/17$ of $170m = 60m$

Now,

$$60m - 50m = 20$$

$$10m = 20$$

$$m = 2$$

Also,

Let the ratio of the total number of orders placed Monday and Friday = $120:100 = 6:5$. Let it be $600n$ and $500n$ respectively.

Monday

The total number of orders placed = $600n$

The number of non-delivered orders = 25% of $600n = 150n$

The number of delivered orders = $(600n - 150n) = 450n$

The number of delivered orders accepted = $5/6$ of $450n = 375n$

The number of delivered orders returned = $1/6$ of $450n = 75n$

Friday

The total number of orders placed = $500n$

The number of non-delivered orders = 30% of $500n = 150n$

The number of delivered orders = $(500n - 150n) = 350n$

The number of delivered orders accepted = $7/10$ of $350n = 245n$

The number of delivered orders returned = $3/10$ of $350n = 105n$

Now,

$$245n - 75n = 102$$

$$170n = 102$$

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$$n = 0.60$$

Also,

Let the total number of orders placed on Wednesday be 100k

So, the total number of non-delivered orders on Wednesday = 10% of 100k = 10k

The number of delivered orders on Wednesday = (100k-10k) = 90k

The number of delivered orders accepted on Wednesday = $\frac{3}{5}$ of 90k = 54k

The number of delivered orders returned on Wednesday = $\frac{2}{5}$ of 90k = 36k

Now,

$$36k - 10k = 39$$

$$26k = 39$$

$$k = 1.50$$

Monday

The total number of orders placed = $600n = (600 \times 0.60) = 360$

The number of non-delivered orders = $150n = (150 \times 0.60) = 90$

The number of delivered orders = $450n = (450 \times 0.60) = 270$

The number of delivered orders accepted = $375n = (375 \times 0.60) = 225$

The number of delivered orders returned = $75n = (75 \times 0.60) = 45$

Tuesday

The total number of orders placed = $100m = (100 \times 2) = 200$

The number of non-delivered orders = $20m = (20 \times 2) = 40$

The number of delivered orders = $80m = (80 \times 2) = 160$

The number of delivered orders accepted = $50m = (50 \times 2) = 100$

The number of delivered orders returned = $30m = (30 \times 2) = 60$

Wednesday

The total number of orders placed = $100k = (100 \times 1.50) = 150$

The total number of non-delivered orders = $10k = (10 \times 1.50) = 15$

The number of delivered orders = $90k = (90 \times 1.50) = 135$

The number of delivered orders accepted = $54k = (54 \times 1.50) = 81$

The number of delivered orders returned = $36k = (36 \times 1.50) = 54$

Thursday

The total number of orders placed = $200m = (200 \times 2) = 400$

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The number of non-delivered orders = $30m = (30 \times 2) = 60$

The number of delivered orders = $170m = (170 \times 2) = 340$

The number of delivered orders accepted = $110m = (110 \times 2) = 220$

The number of delivered orders returned = $60m = (60 \times 2) = 120$

Friday

The total number of orders placed = $500n = (500 \times 0.60) = 300$

The number of non-delivered orders = $150n = (150 \times 0.60) = 90$

The number of delivered orders = $350n = (350 \times 0.60) = 210$

The number of delivered orders accepted = $245n = (245 \times 0.60) = 147$

The number of delivered orders returned = $105n = (105 \times 0.60) = 63$

Days	The total number of orders placed	The number of orders delivered	The number of non-delivered orders	The number of delivered orders accepted	The number of delivered orders returned
Monday	360	270	90	225	45
Tuesday	200	160	40	100	60
Wednesday	150	135	15	81	54
Thursday	400	340	60	220	120
Friday	300	210	90	147	63
Total	1410	1115	295	773	342

Answer: C

The total number of orders placed on Tuesday = 200

Saturday

The total number of orders placed = 90% of 200 = 180

The number of non-delivered orders = $4Z$

So, the number of delivered orders = $(180 - 4Z)$

The number of delivered orders accepted = $\frac{3}{4}$ of $(180 - 4Z) = (135 - 3Z)$

The number of delivered orders returned = $\frac{1}{4}$ of $(180 - 4Z) = (45 - Z)$

Now,

$$(45 - Z) - 4Z = 20$$

$$45 - 5Z = 20$$

$$5Z = 25$$

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$$Z = 5$$

Therefore, the value of $(Z^2+5) = (5^2+5) = (25+5) = 30$.

Hence, the correct answer is option C.

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